

ECEN721: Optical Interconnects Circuits and Systems Spring 2026

Lecture 14: RF Silicon Photonic Receiver Front-Ends



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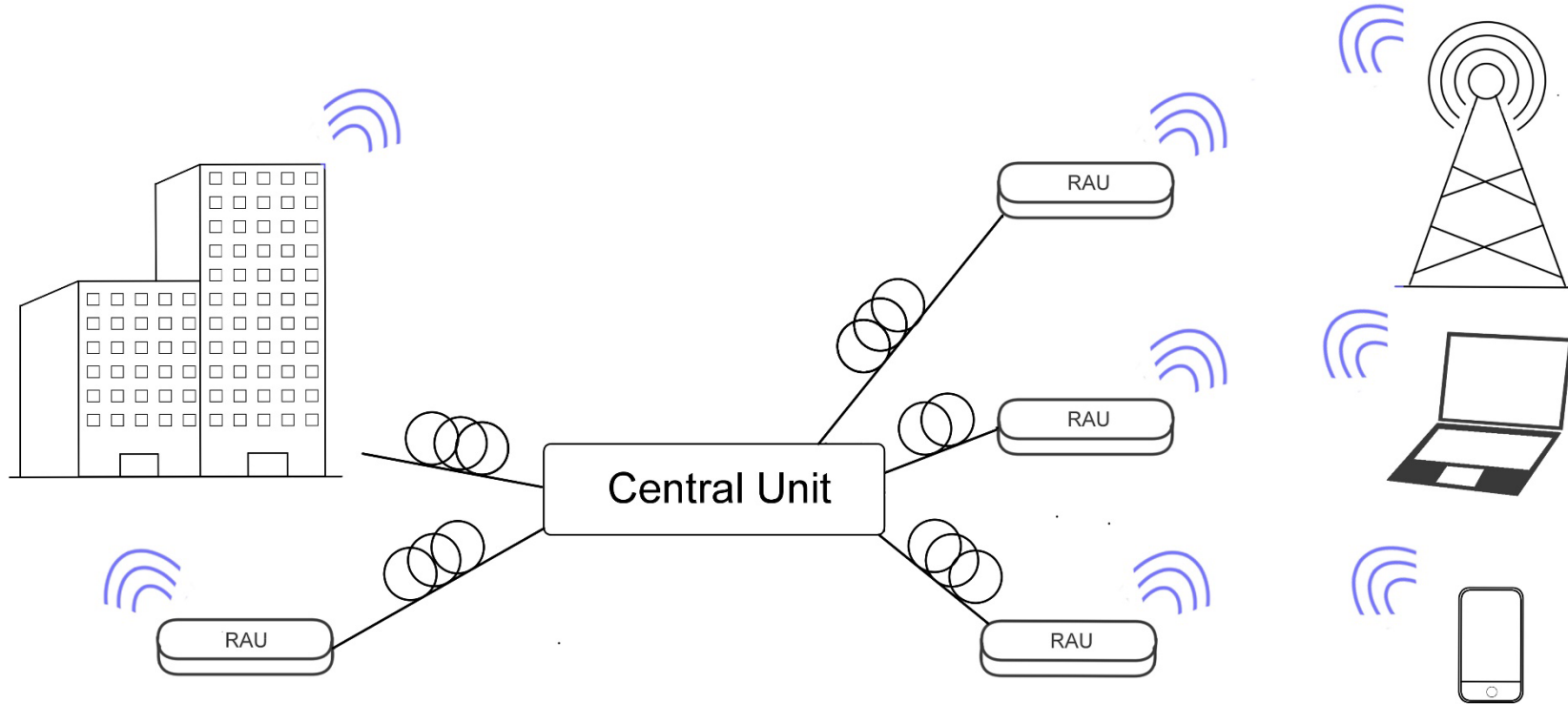
Announcements

- Project Preliminary Report (HW4) Due Apr 14
- Project Final Report Due Apr 28
- Exam 2 May 4 (1PM-3PM)

Outline

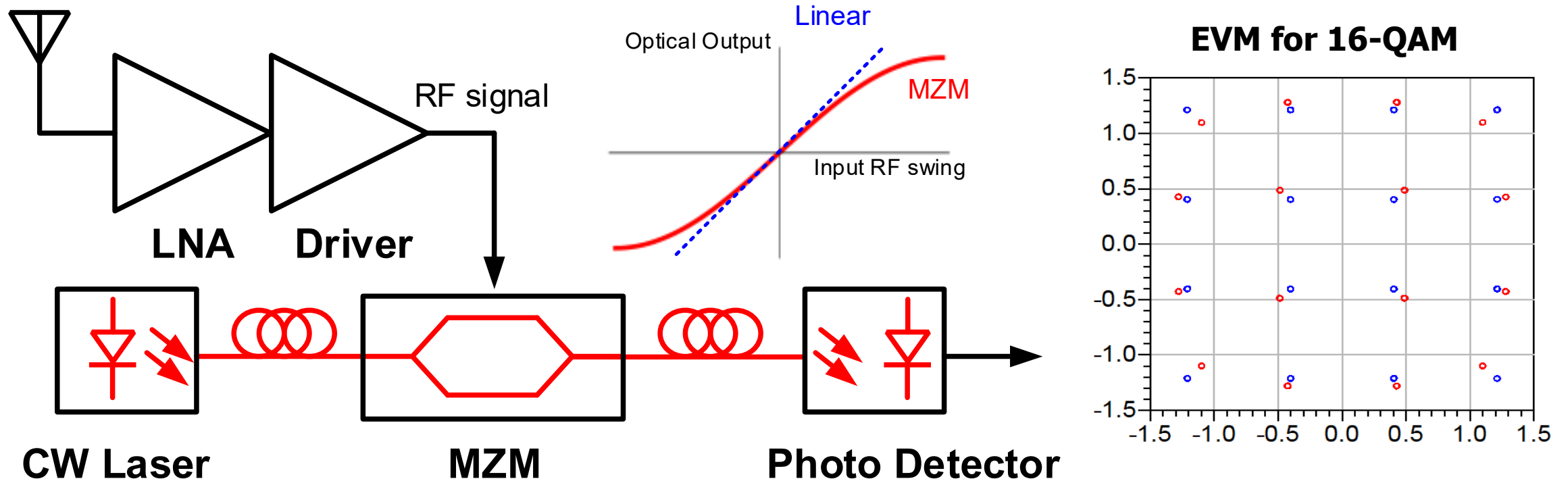
- A Power-Efficient 20–35 GHz MZM Driver with Programmable Linearizer in 28nm CMOS
- A 16-32GHz RF Silicon Photonic Receiver with 22nm FD-SOI CMOS Driver
- A 15-25GHz RF Photonic Front-End with 22nm CMOS Dual-Differential Driver and Silicon Traveling-Wave Mach-Zehnder Modulator

Radio-Over-Fiber Systems



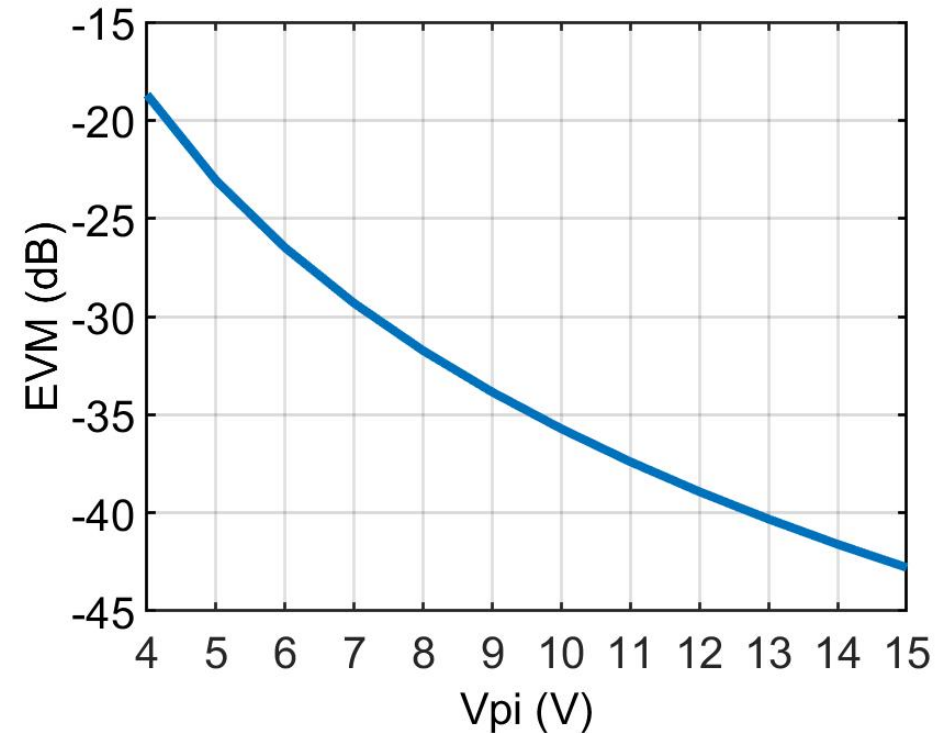
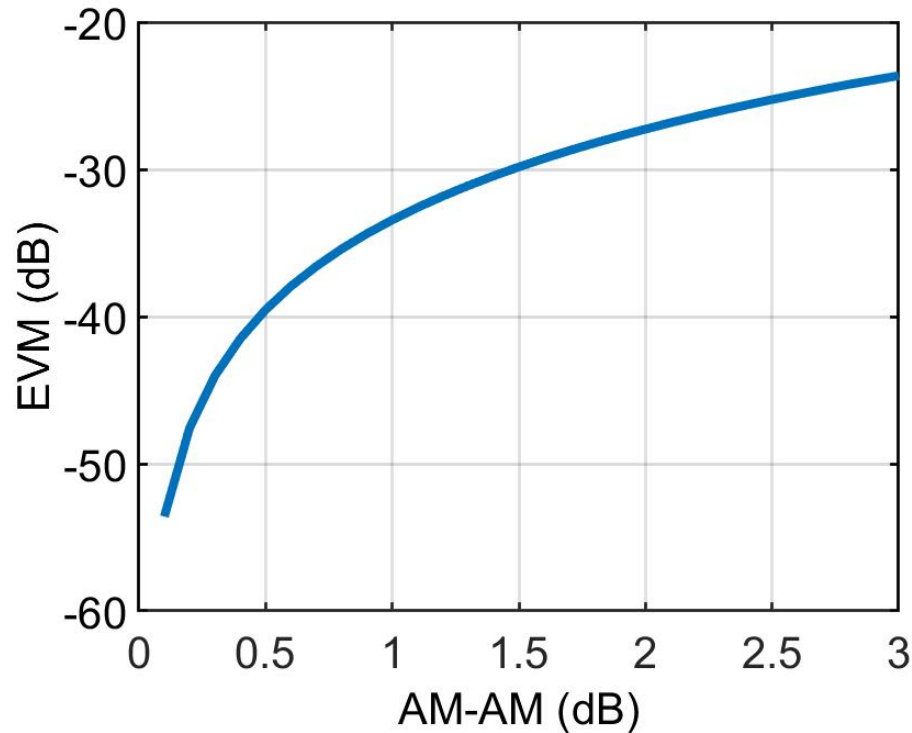
- Remote antenna units
- Indoor wireless communications
- 5G cellular communications

Nonlinearity in RoF Systems



- MZM cosine transfer function
 - Major source of nonlinearity
 - AM-AM compression

16-QAM EVM vs AM-AM

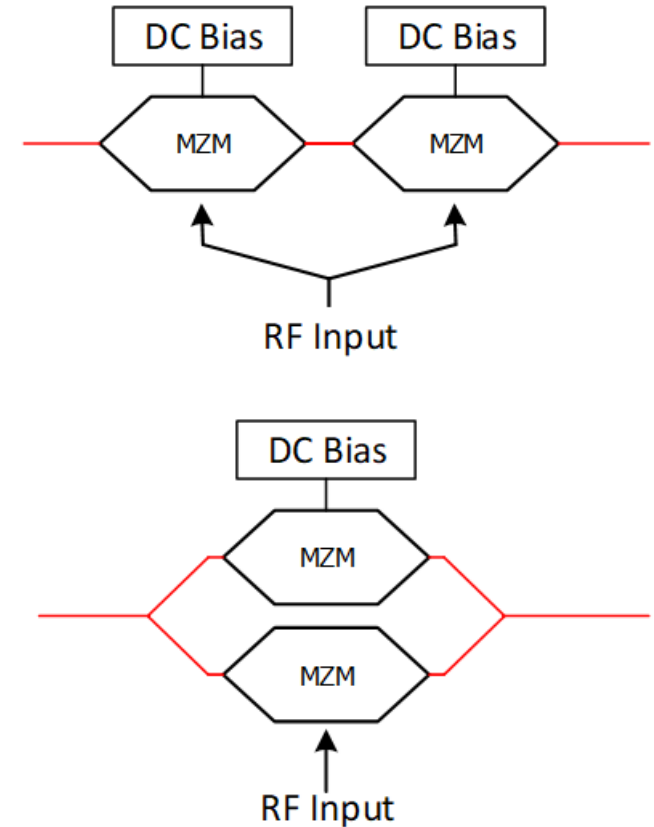


- Assuming ideal cosine model for MZM
- EVM degrades when AM-AM compression increases
- Lower V_{π} MZM has higher gain, but more nonlinearity

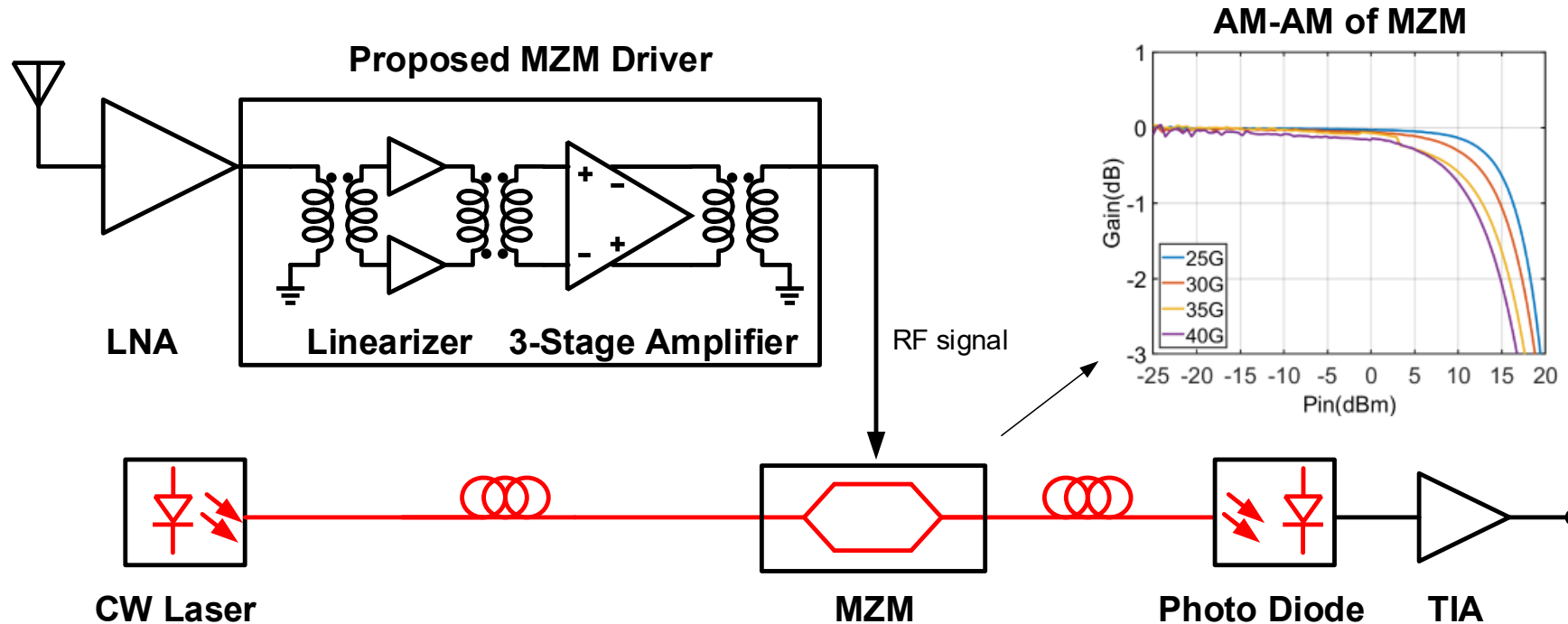
MZM Linearization Approaches

- Optical domain linearization occupies large PIC area
- Electrical domain linearization approaches
 - Arcsine function from square law of transistor
 - Polynomial predistortion
 - IM3 injection
 - Diode-based predistortion
- Proposed programmable linearizer compensates AM-AM and is highly tunable to generate predistortion

Optical-Domain Linearization



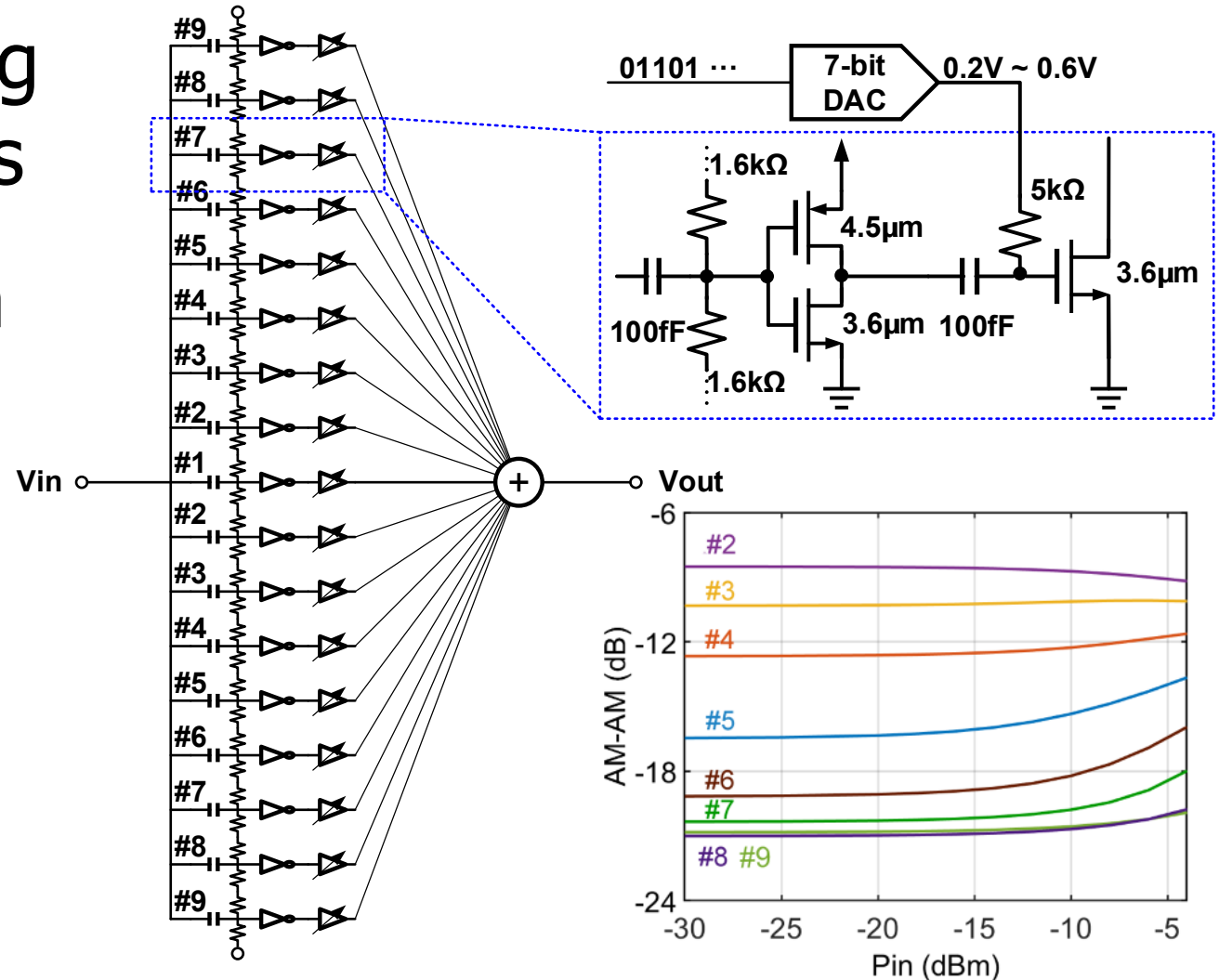
MZM Driver Architecture



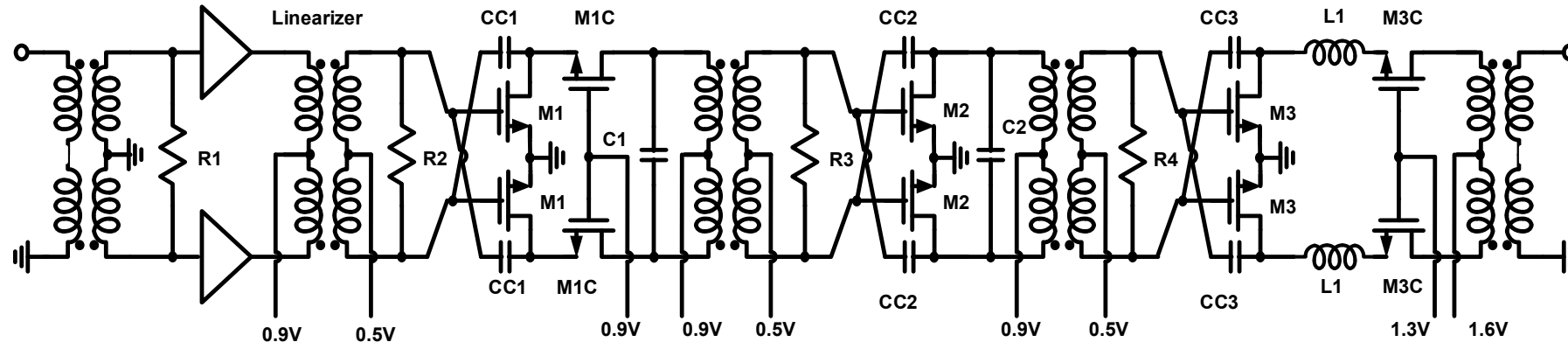
- Programmable inverter-based amplifier linearizer generates predistortion to compensate MZM nonlinearity
- 3-stage amplifier provides gain to deliver 12dBm linear output power to drive MZM

Inverter-Based Amplifier Linearizer

- 17 unit segments consisting of inverter-based amplifiers
- Segments 1-3 provide gain without major expansion
- Segments 4-9 provide signal expansion at progressively larger input power levels



CMOS Driver



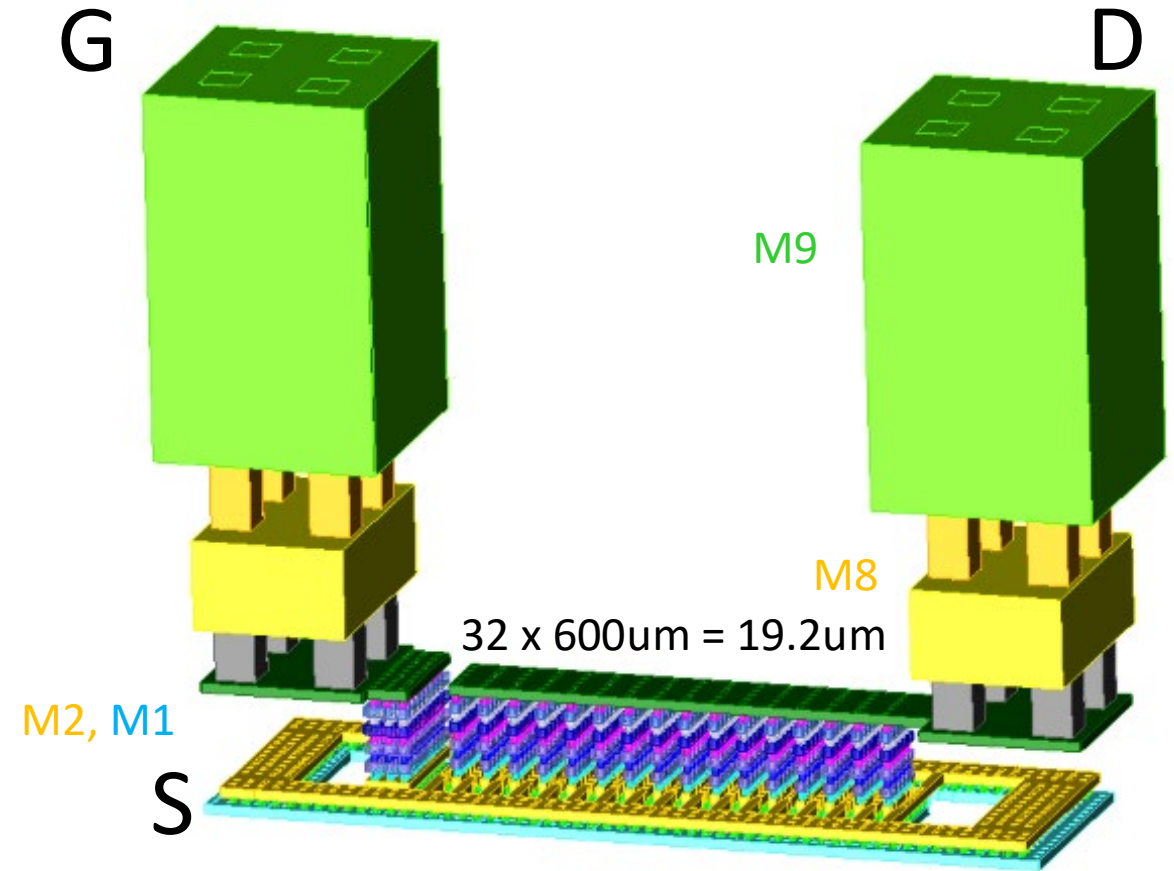
R1	R2	CC1	M1	M1C	C1	R3
160Ω	160Ω	30fF	19.2μm x 5	19.2μm x 5	60fF	50Ω

CC2	M2	C2	R4	CC3	L1	M3	M3C
20fF	19.2μm x 10	50fF	50Ω	40fF	60pH	19.2μm x 24	19.2μm x 18

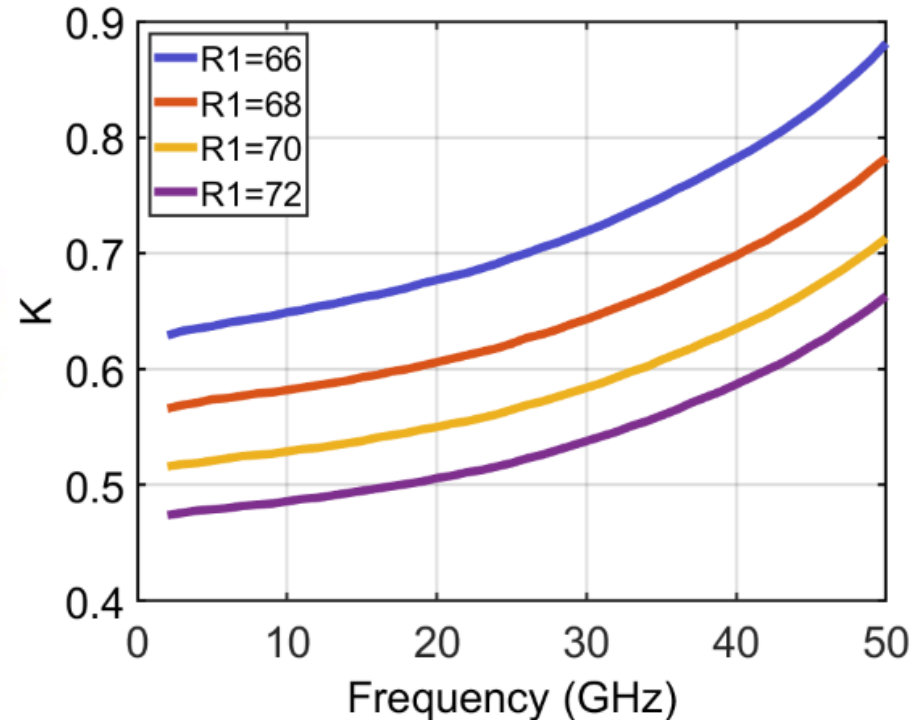
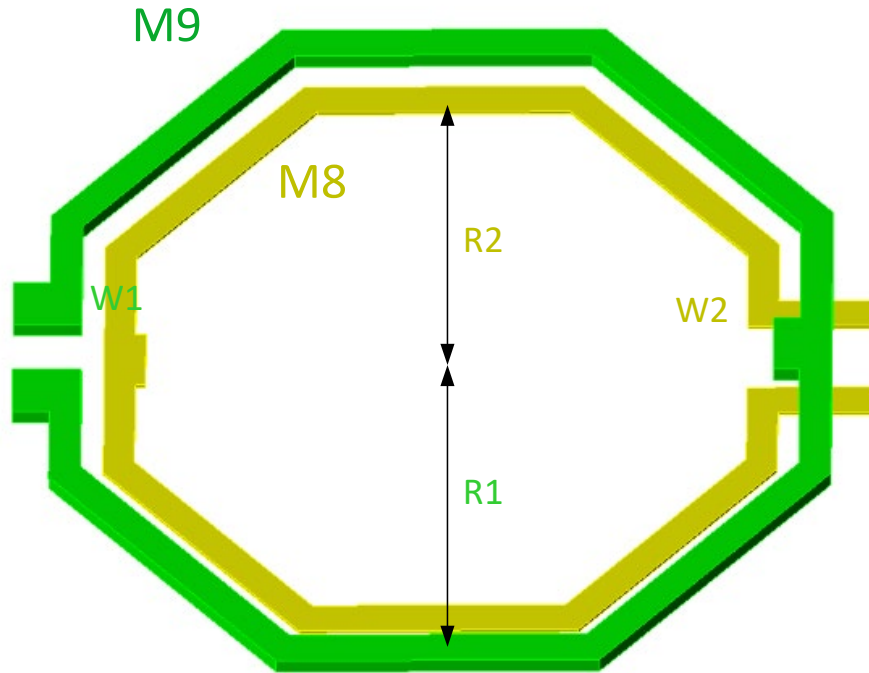
- 3 pseudo-differential common-source amplifier stages
- Capacitive neutralization improves reverse isolation
- Cascode structures in first stage improves stability and in last stage allows for operation with a higher supply
- Inter-stage and output-stage matching implemented with symmetrical magnetically coupled resonator technique

28nm CMOS Unit Cell Layout

- Unit cell methodology allows for easy scaling of each amplifier stage
 - 32 600nm width fingers
 - Parasitics minimized with higher metal routing for drain connection
 - Gate resistance reduced with double-sided connection
 - Source impedance reduced with stacked M1/2 layers



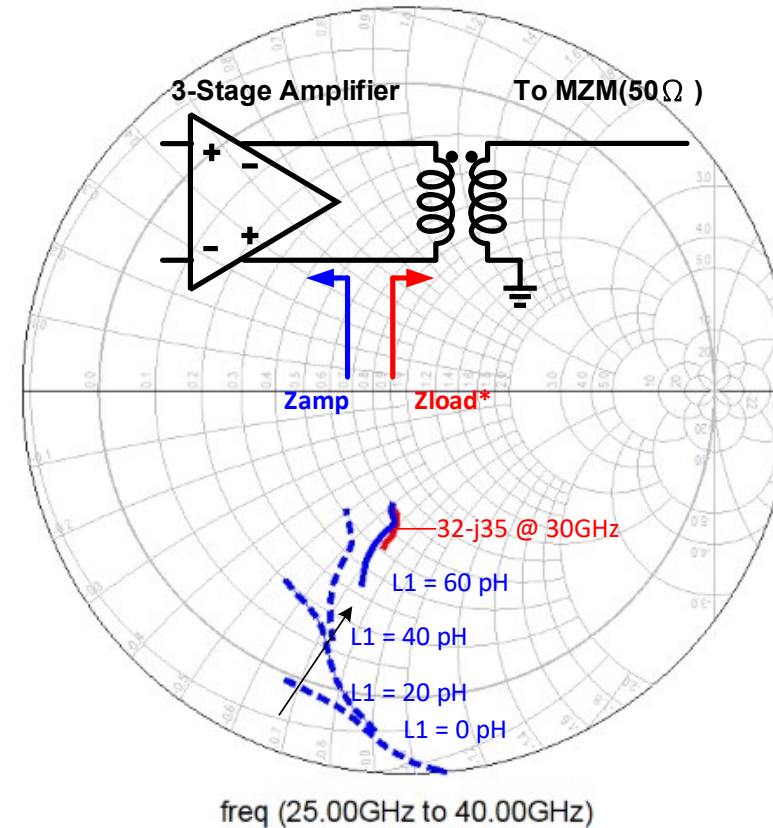
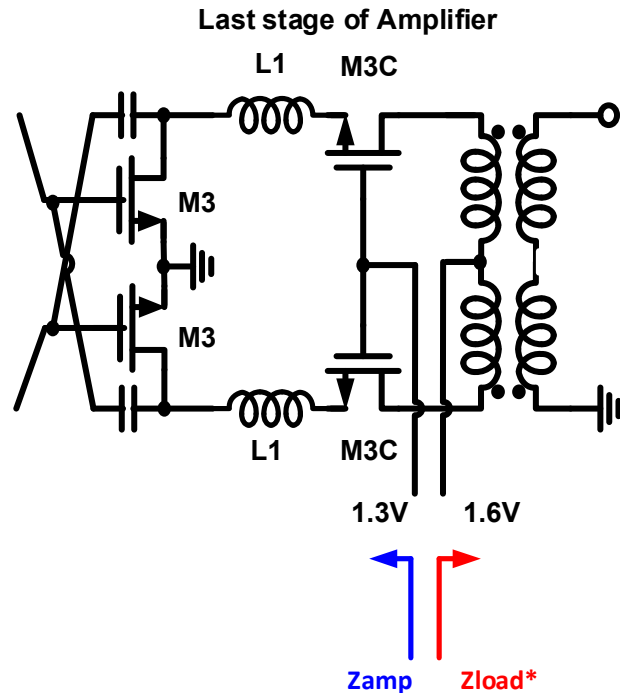
Mutual-Coupling Resonator (MCR)



- Adjusting coupling factor by tuning radius
- Requirement for zero gain ripple

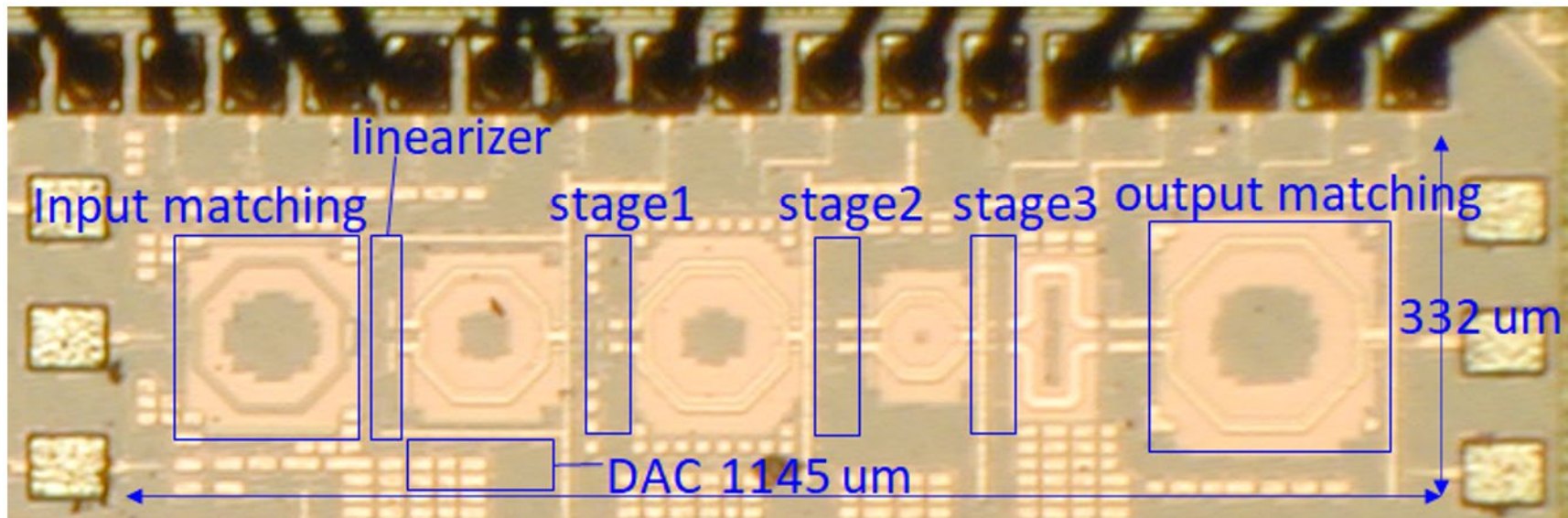
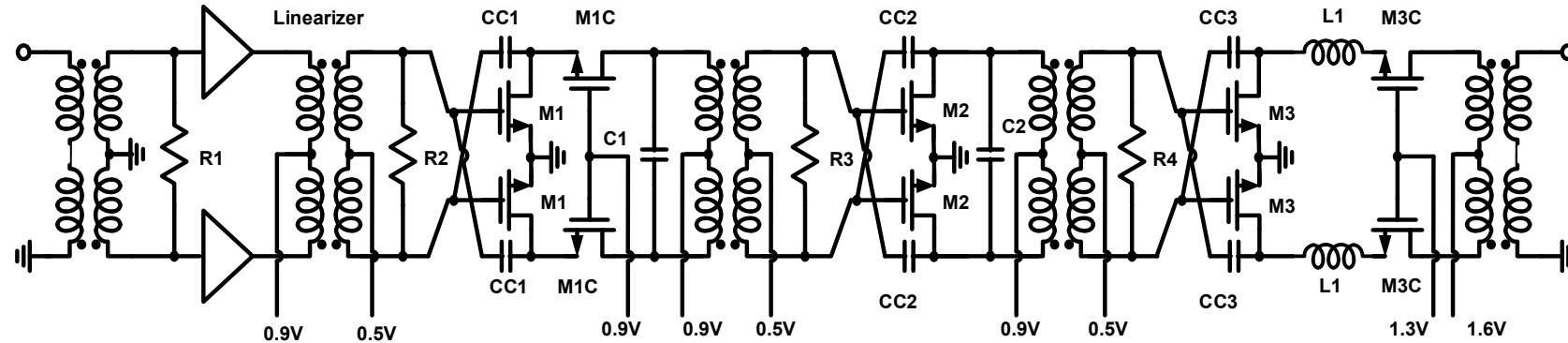
$$k'^2 (Q^2 + 1) = 1$$

Output Matching



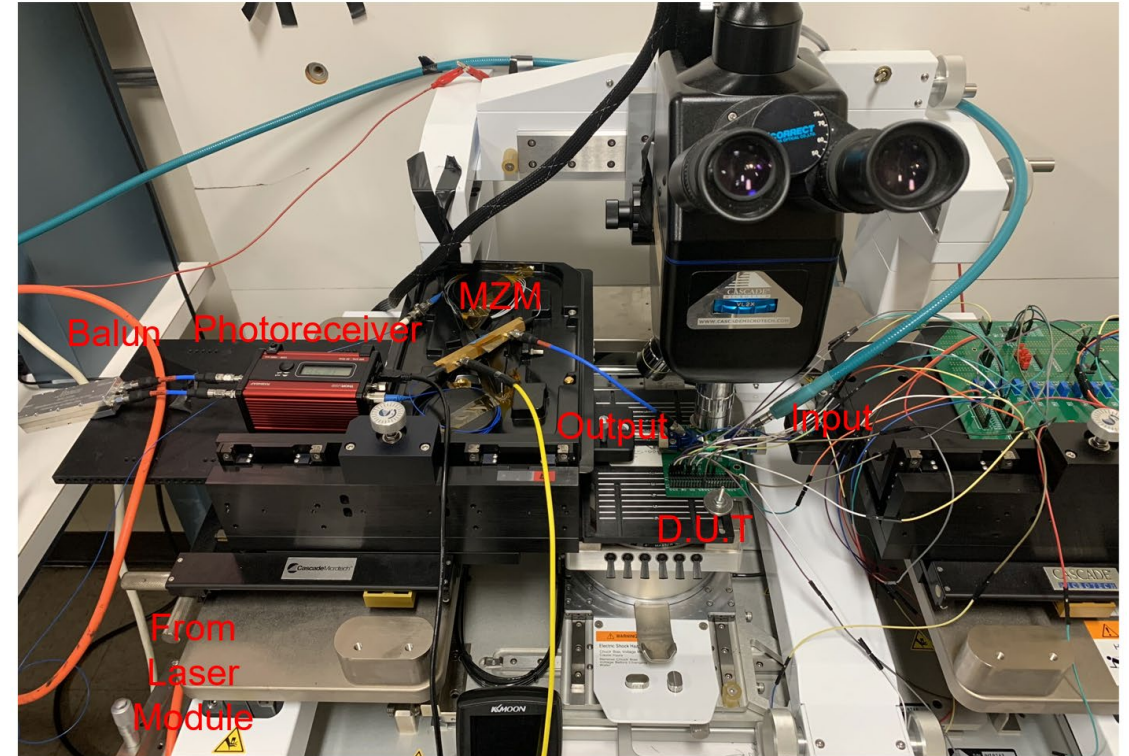
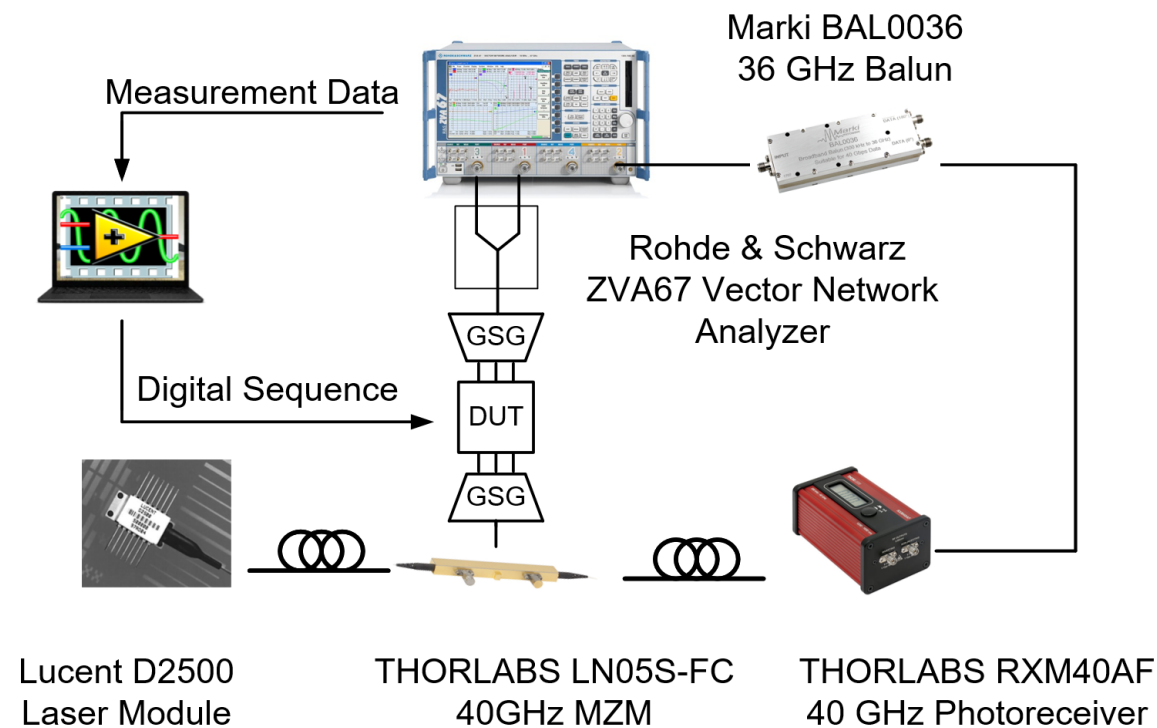
- Inductor L1 adopted to tune the impedance Z_{amp}
- Conjugate matching for power delivery and output return loss

28nm CMOS Prototype



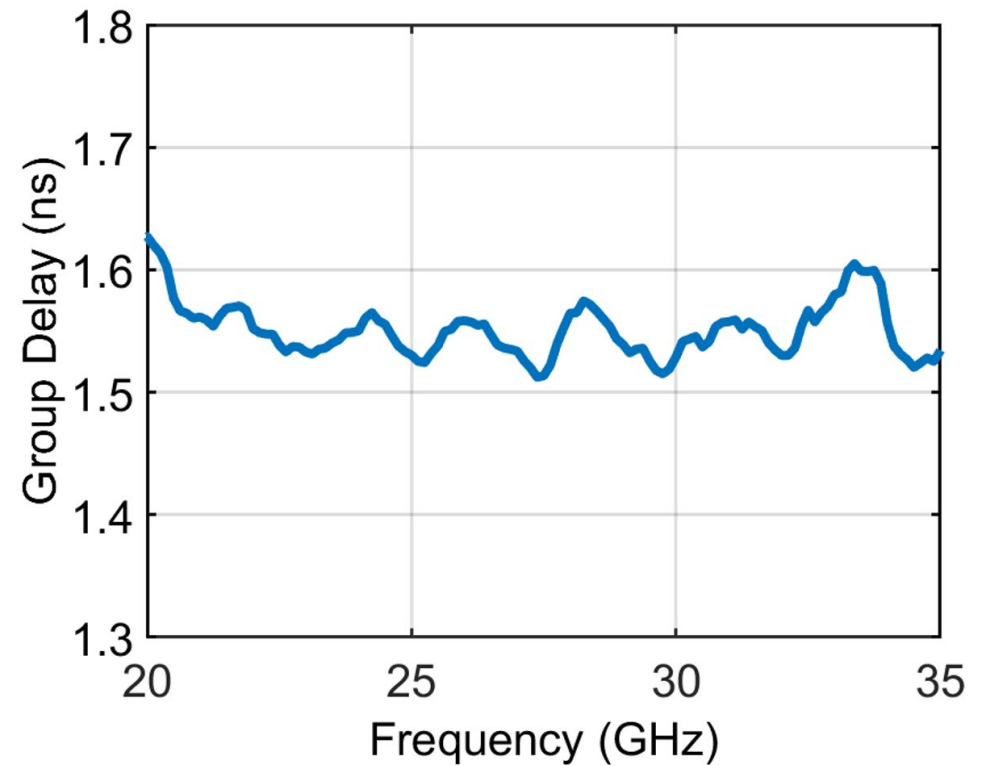
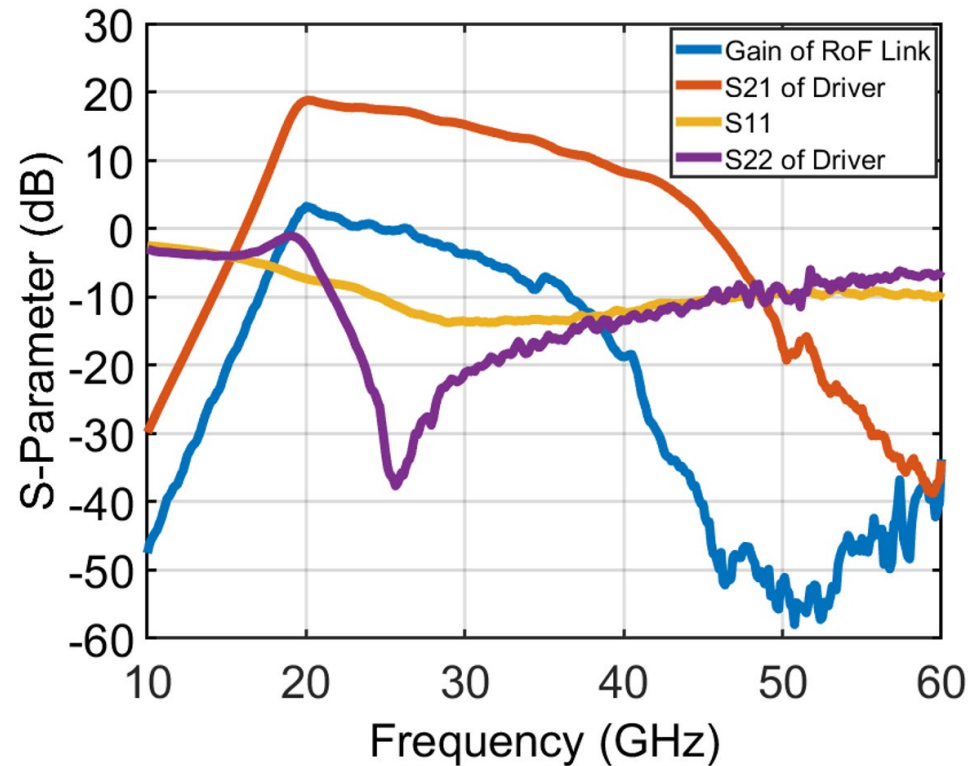
- GSG probe pads for high-speed input and output
- DC supplies and serial control signals applied via wirebonds to PCB

Test Setup



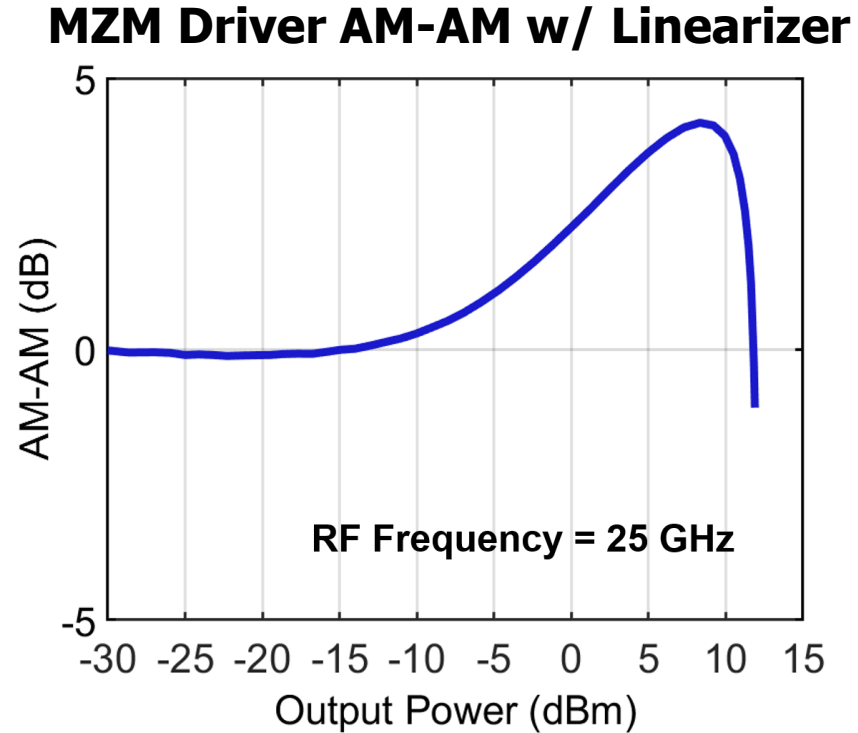
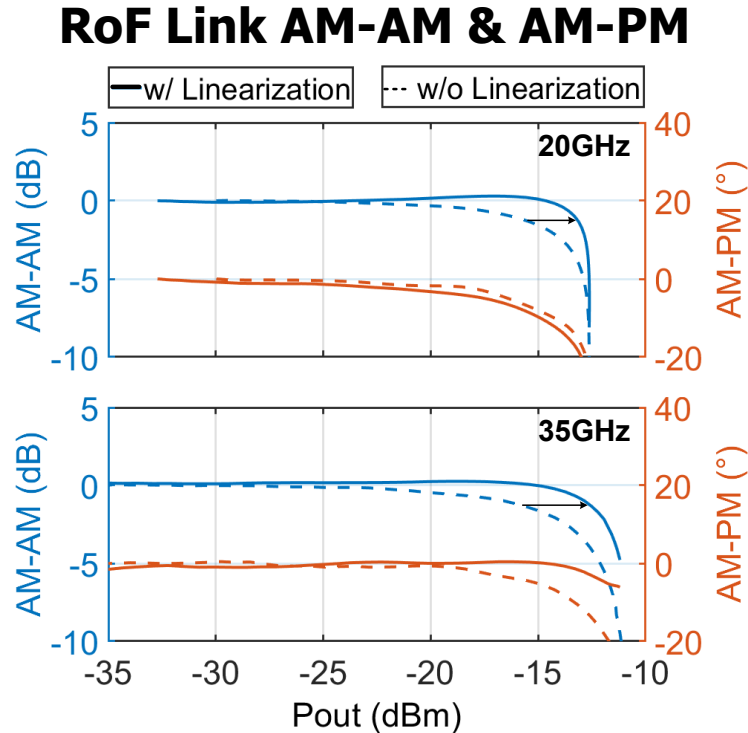
- 40GHz LiNbO3-MZM with 7V RF V_{π} at 30GHz
- MZM biased at quadrature point

S-Parameters & Group Delay



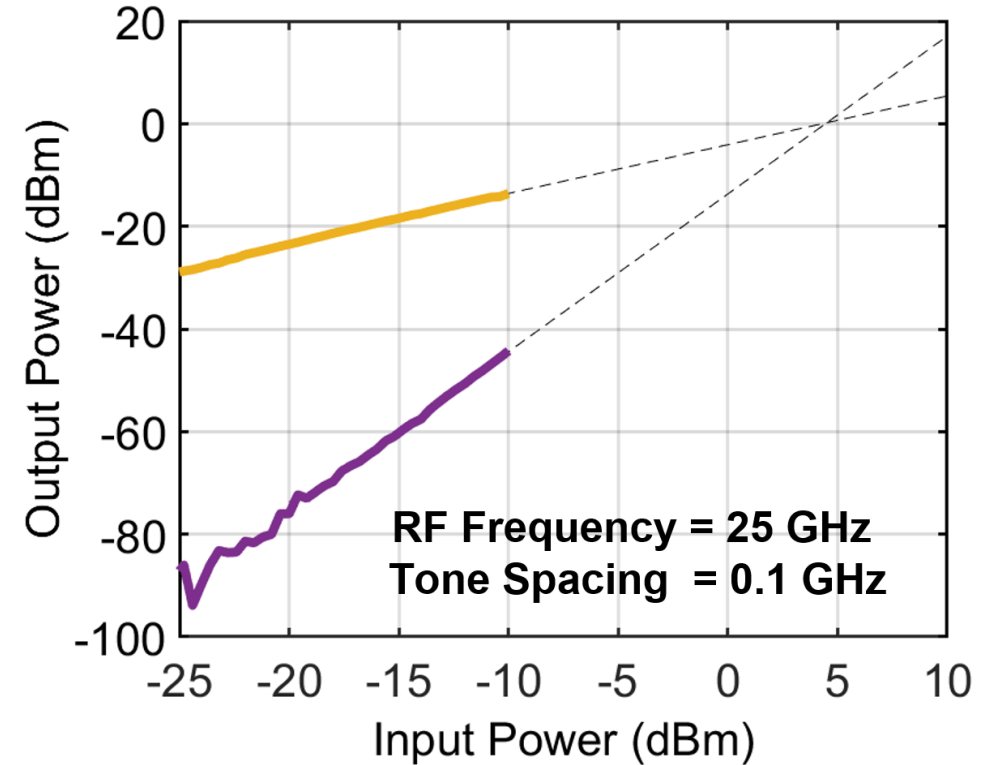
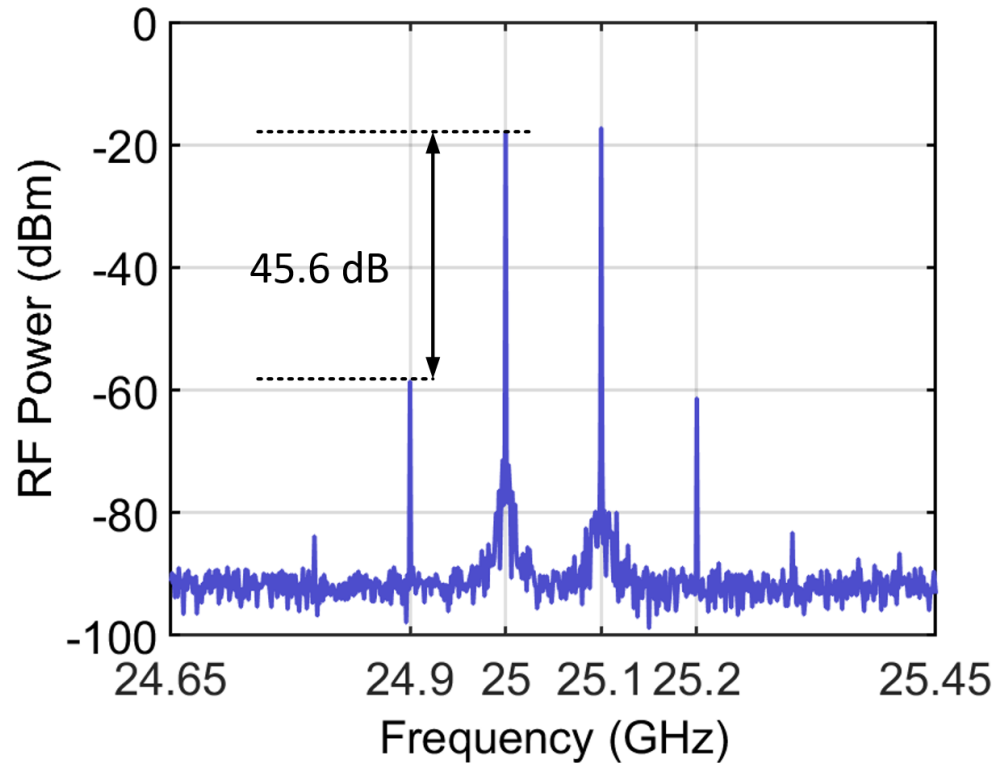
- 20-35GHz 3dB bandwidth with max 18dB gain
- Group delay variation of the entire RoF link is <115ps within the 20-35 GHz bandwidth

AM-AM Compensation



- Activating linearizer allows for 3dB OP1dB extension
- Driver delivers 12dBm output power with expansive response that compensates MZM compression

Two-Tone Measurements



- 4.1dBm IIP3 for entire RoF link

Summary Table

References	[Hosseinzadeh RFIC 2019]	[Sadhwani JLT 2003]	[Okyere TX Symp 2017]	This Work
Technology	Si-SiGe	CMOS 180nm	65nm CMOS	28nm CMOS
Frequency	0.5-20GHz	0.28GHz	1GHz	20-35GHz
Power Consumption	1700mW*	162mW	49.2mW	180mW
Max Voltage Swing	$2V_{pp}$	N/A	N/A	$2.5V_{pp}$
IIP3	22dBm	6.8dBm	N/A	4.1dBm
Supply Voltage	2.5/3.3V	1.8V	N/A	0.9/1.6V
Power Efficiency**	11.76 GHz/W	1.73GHz/W	N/A	194.44GHz/W
Technique	IM3 Injection	Adaptive Predistoriton	Polynomial Predistortion	Predistortion

*Total power of 4 stages

**Power Efficiency = Max Frequency / Power Consumption

MZM Driver w/ Programmable Linearizer Summary

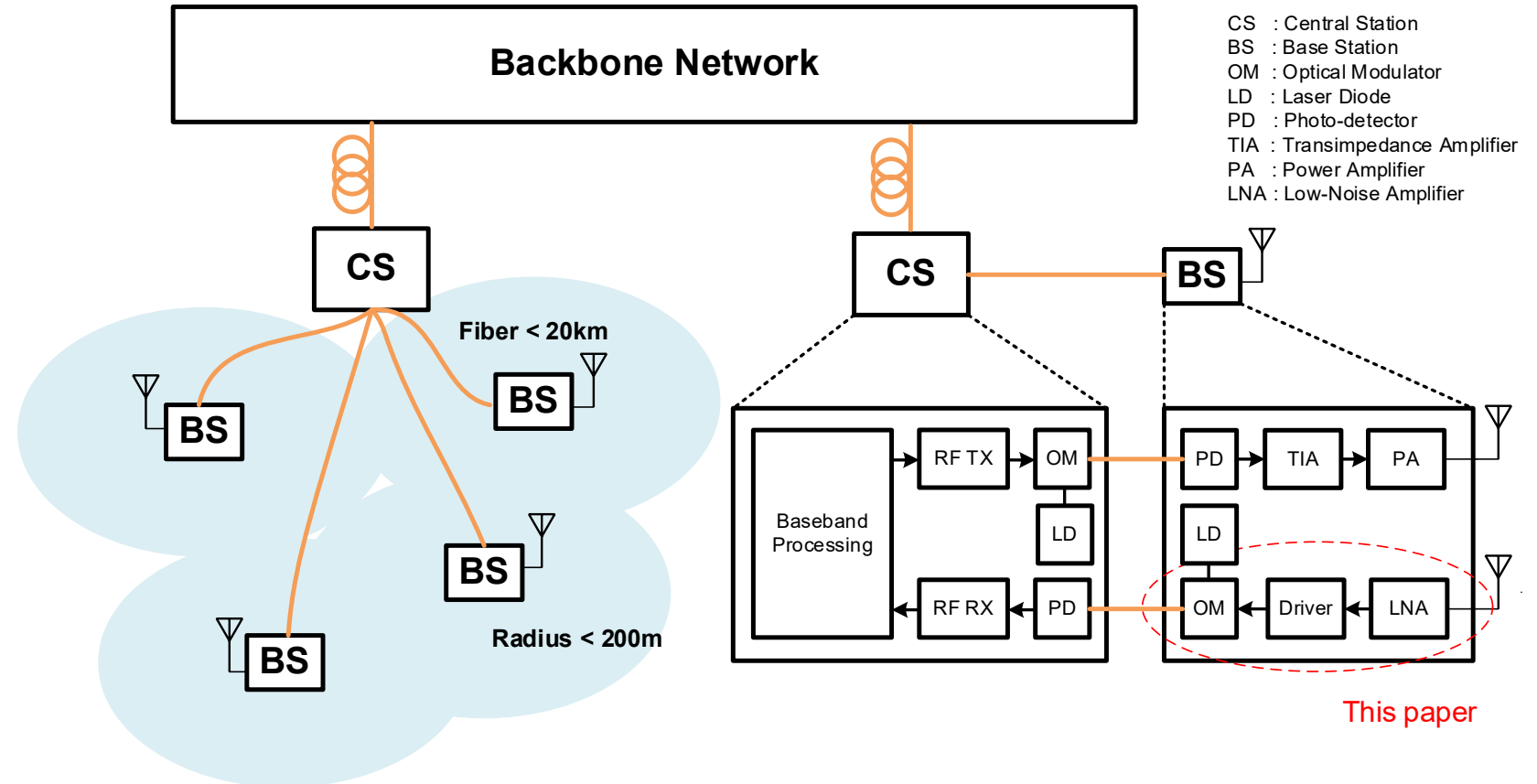
- Power-efficient 28nm CMOS MZM driver for an external MZM achieves 20-35GHz bandwidth and $2.5V_{pp}$ output swing
- Programmable linearizer extends OP1dB by 3dB and has the flexibility to support different MZM types

Outline

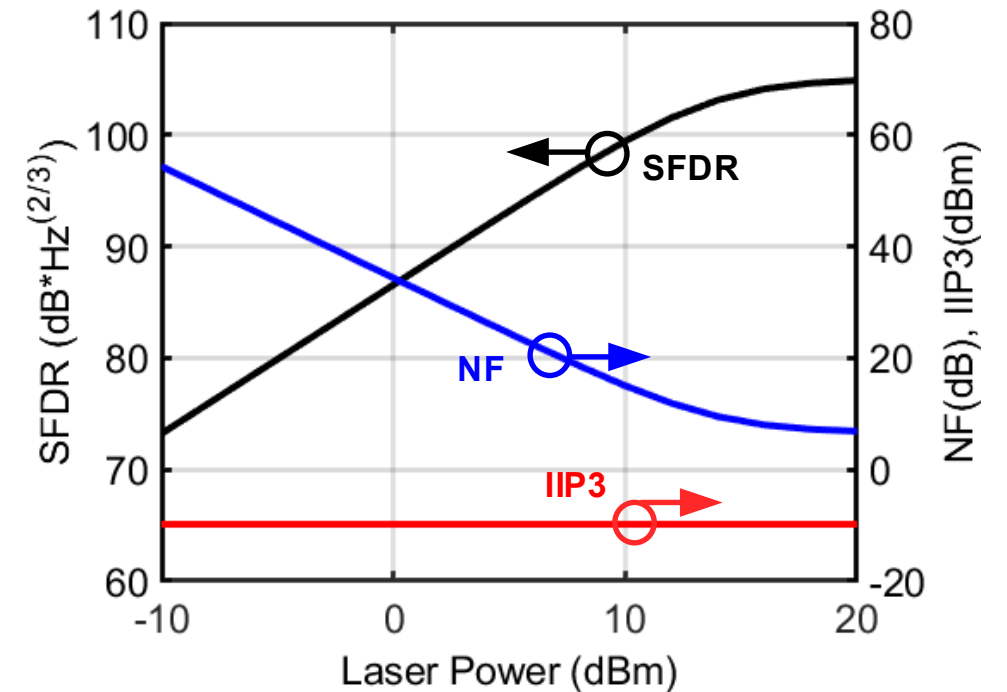
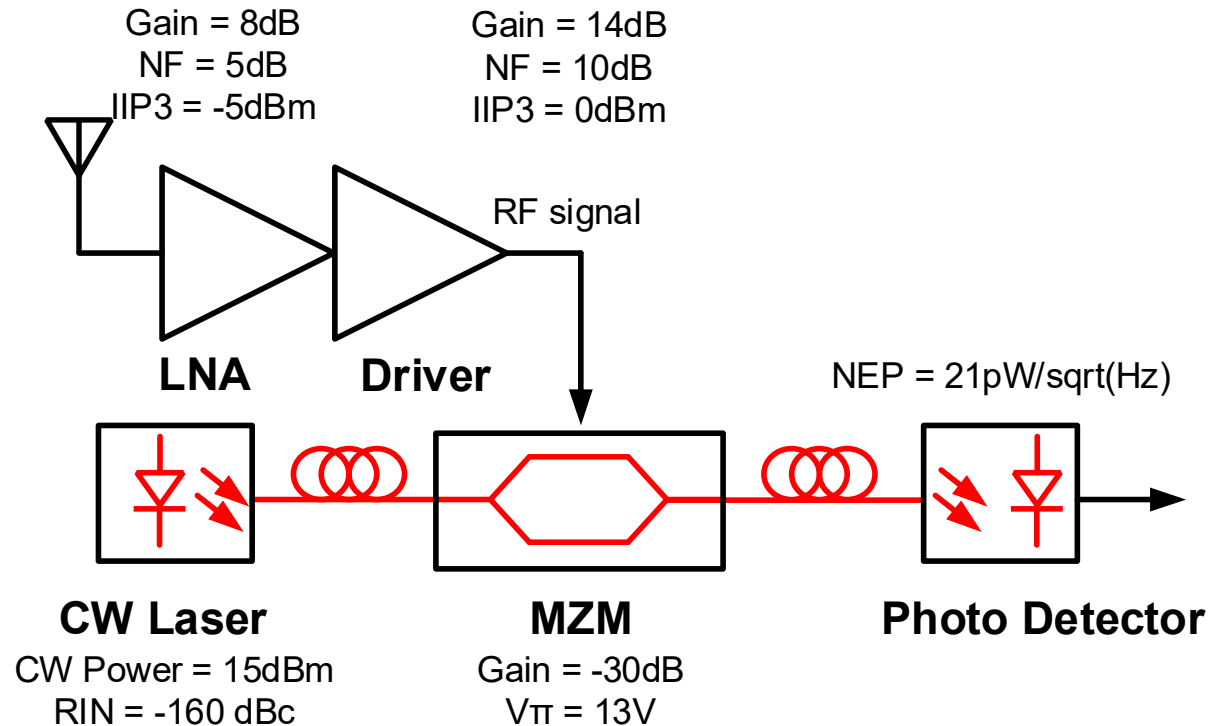
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Radio-over-fiber System

- Indoor, wireless, road vehicle communication
- Central station controls numerous base station remote-antenna-units (RAUs)
- Compact, linear, efficient, low-cost BS units required



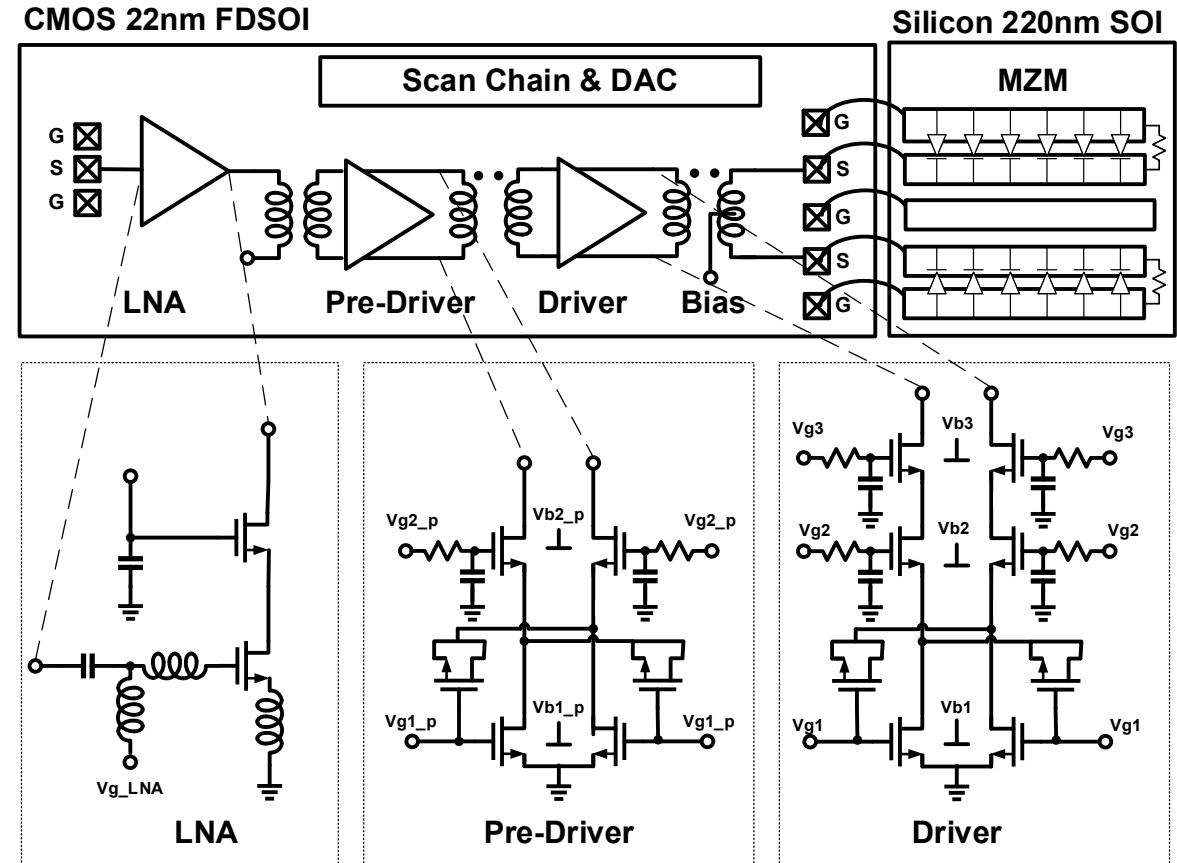
System Analysis



- Electrical amplifier helps to improve SFDR and minimum detectable power

Overall Architecture

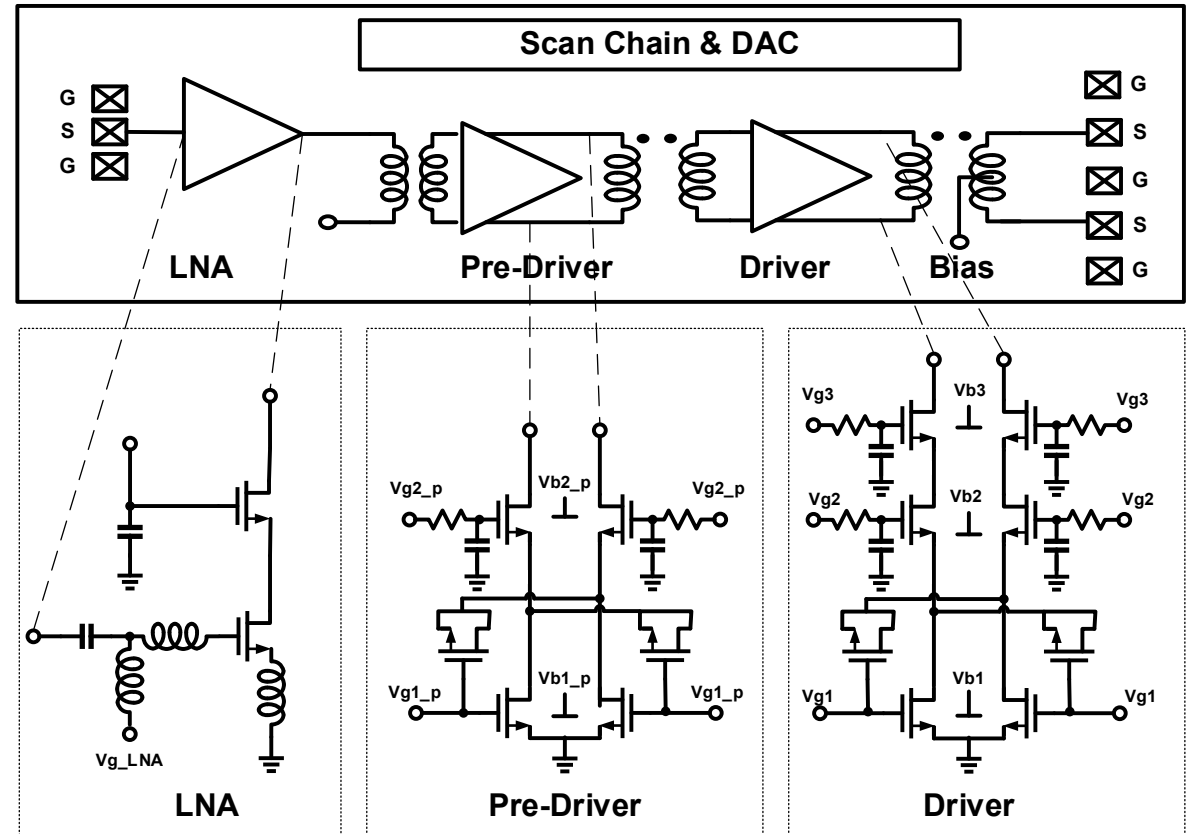
- Electrical low-noise, high-power driver in CMOS 22nm FDSOI
- Differential traveling-wave MZM in Silicon 220nm SOI
- GSGSG configuration bondwire interface
- Reverse-bias voltage fed from CMOS chip



Driver Design

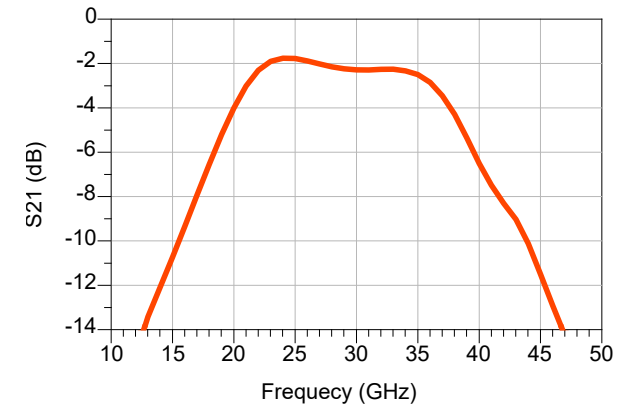
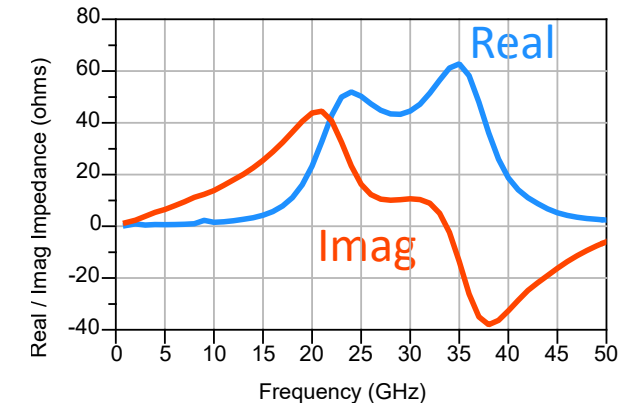
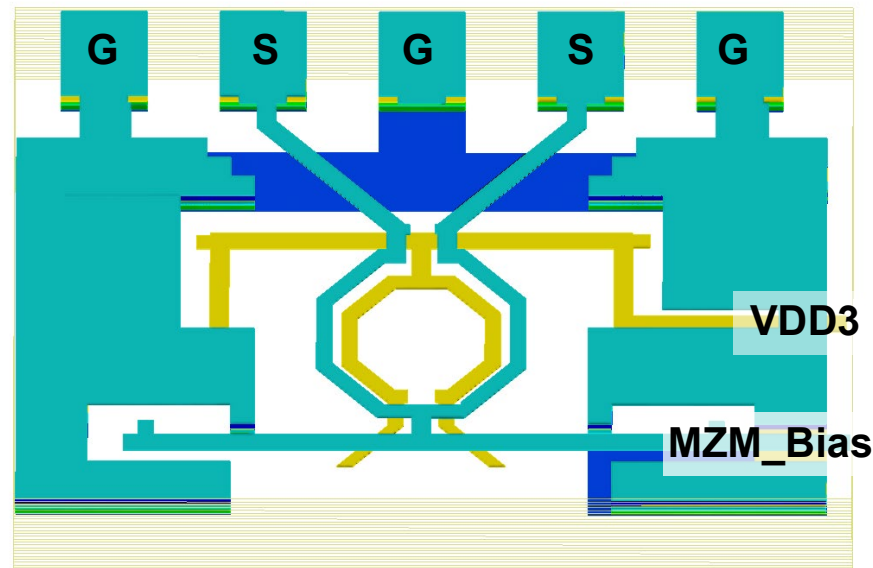
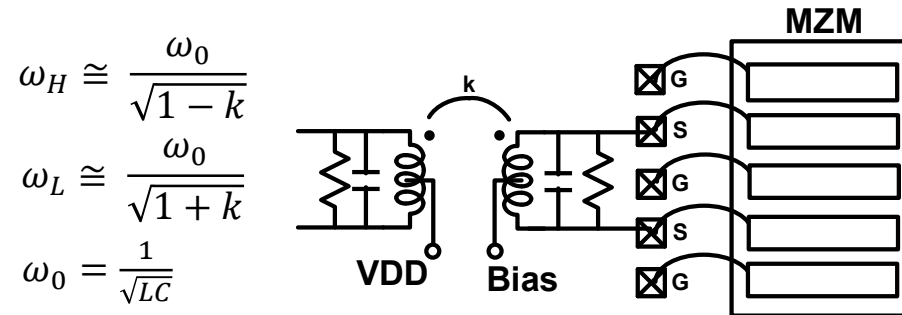
- One-stage cascode LNA
- Stack-FET topology for higher supply driver operation
- Capacitive neutralization provides isolation
- Gate bias voltages generated by on-chip DACs controlled by scan chain

CMOS 22nm FDSOI



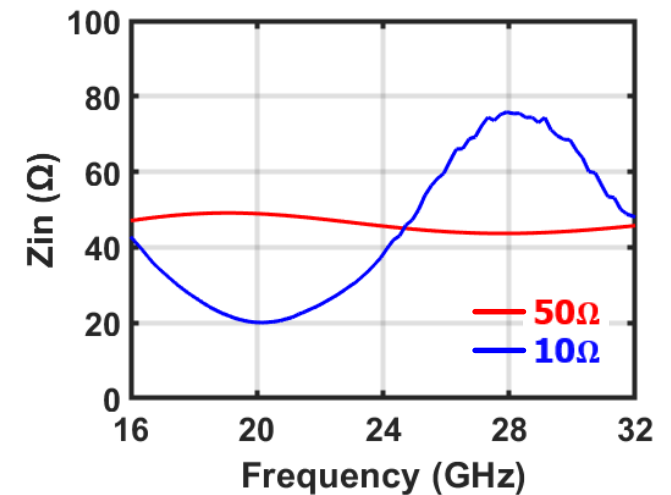
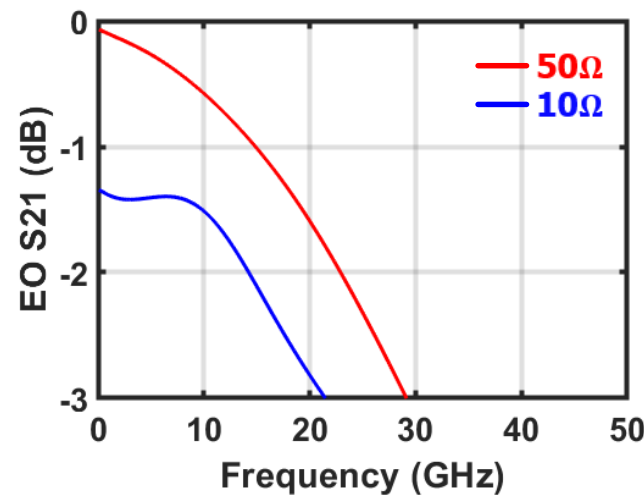
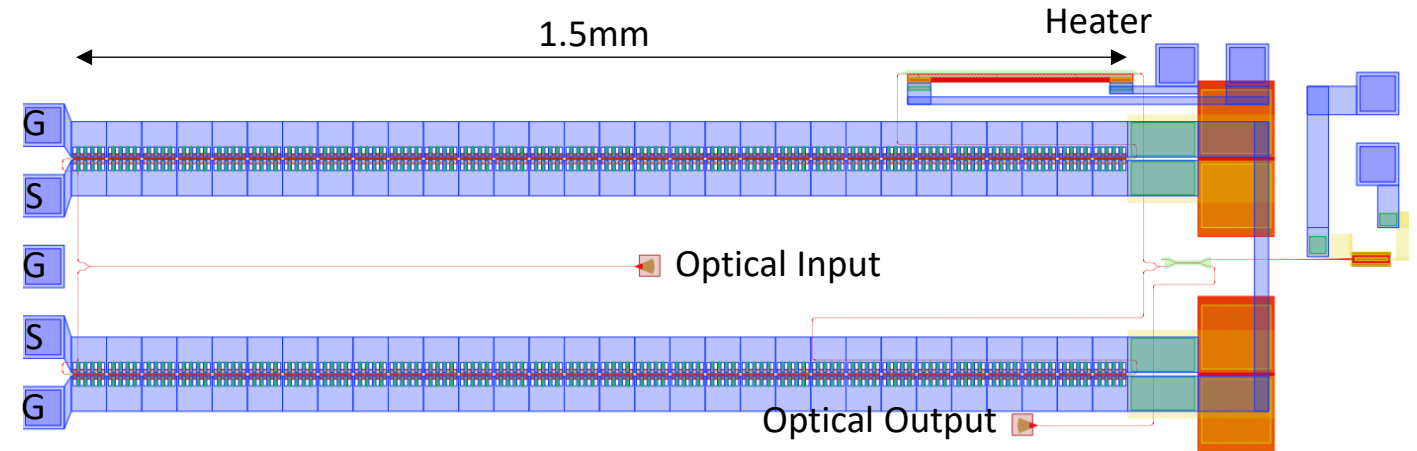
Wideband Design Techniques (MCR)

- Magnetically-coupled resonator (MCR) generates two poles
- Design considerations
 - Bondwire inductance
 - Pad capacitance
 - Transistor parasitics
 - MZM characteristic impedance
 - Routing inductance



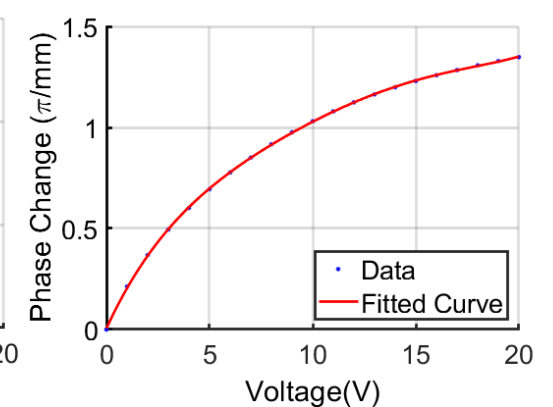
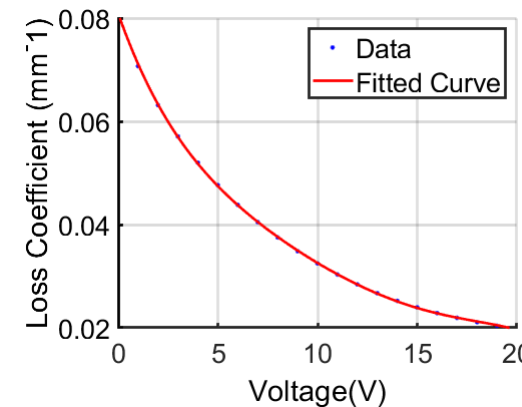
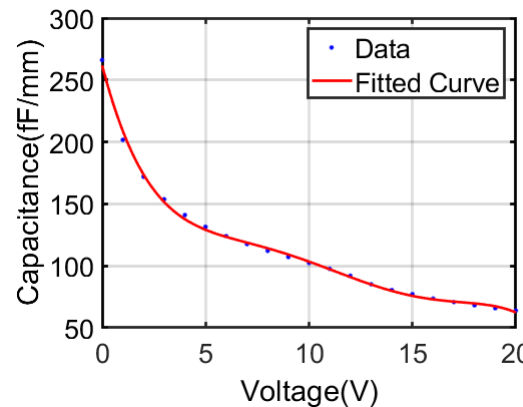
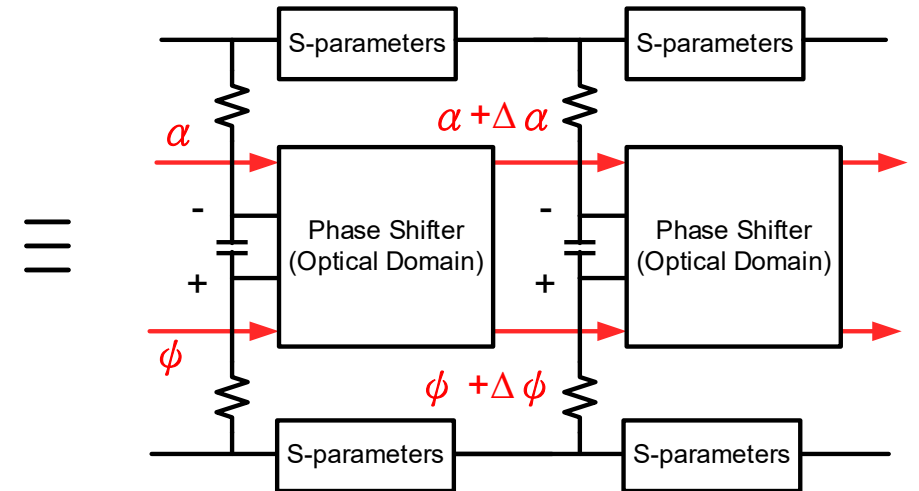
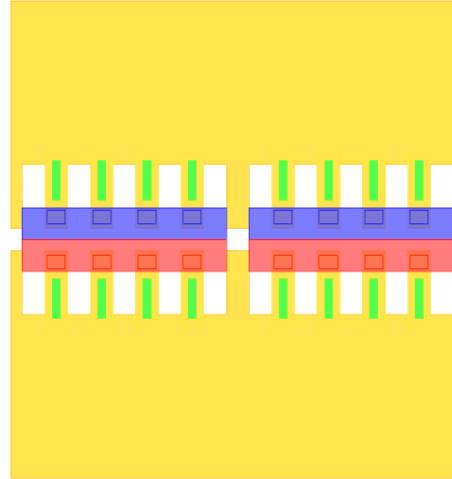
MZM Design

- Slow-wave structure for matching RF index and optical index
- Heater for bias control
- Simulated 29GHz bandwidth with planned 50 Ω termination resistance
- Layout error caused 10 Ω termination mismatch and bandwidth reduction to 22GHz

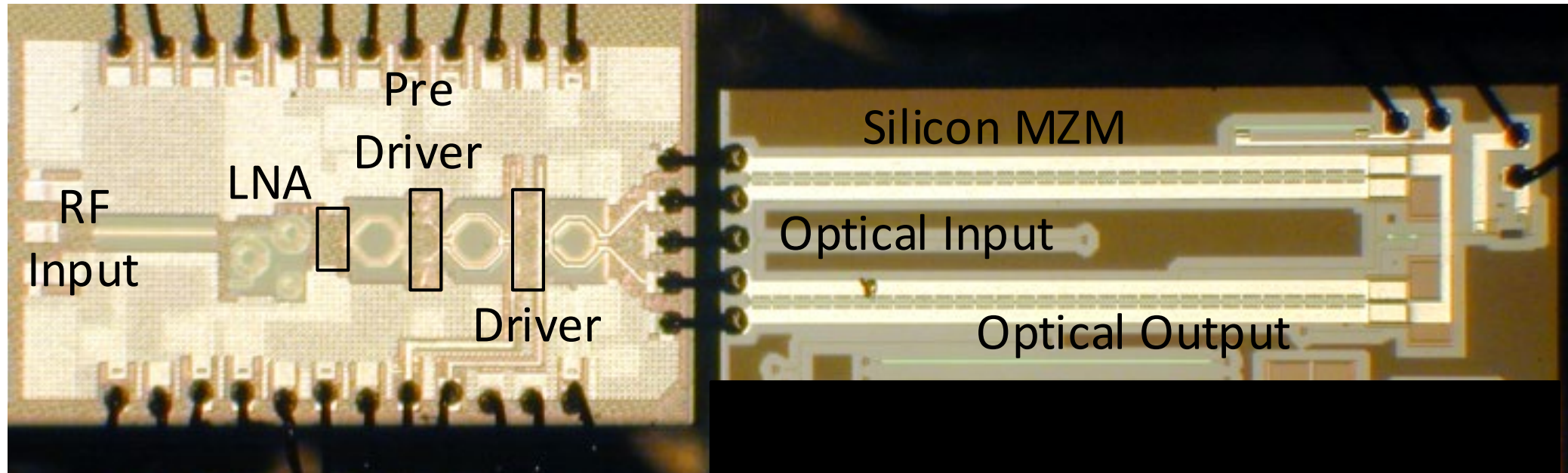


MZM Verilog-A Model

- Electrical model utilizes extracted S-parameters
- Verilog-A used for the optical model
- Complete model captures voltage-dependent junction capacitance, optical loss and phase shift

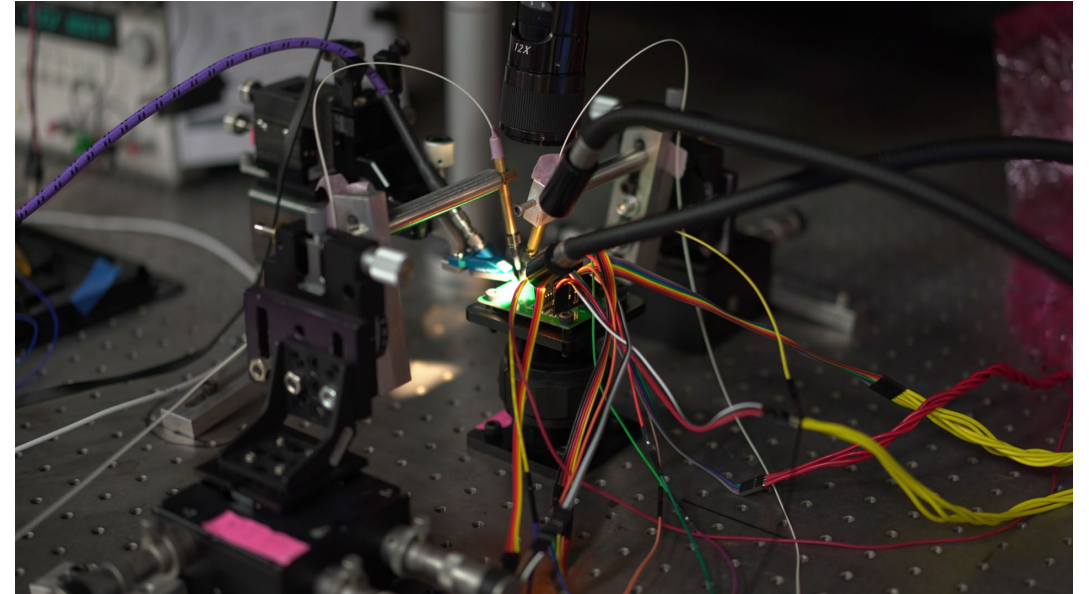
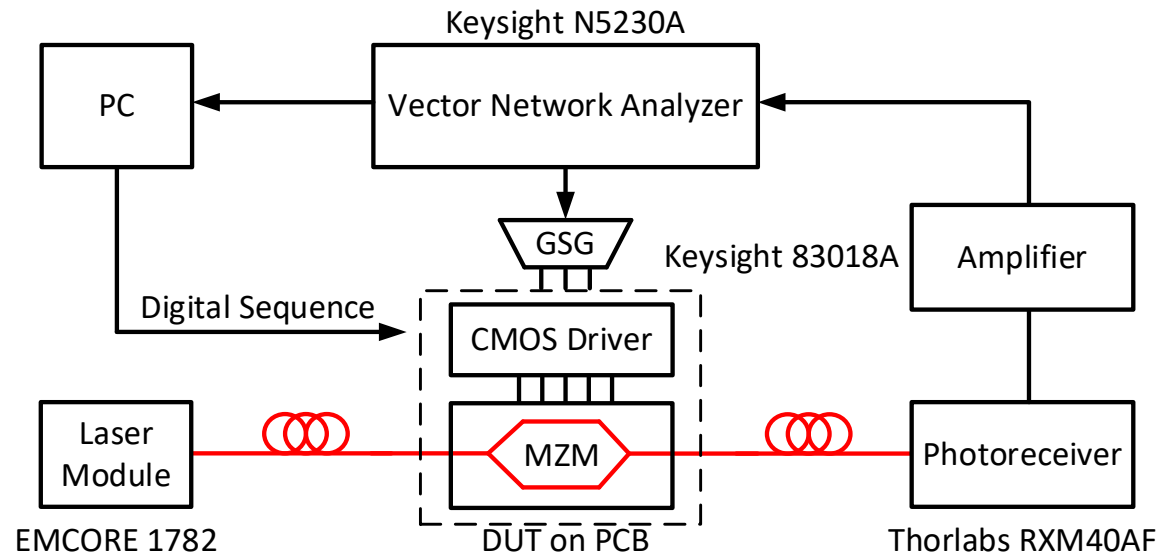


Implementation



- Two chips have the same height to minimize bondwire inductances
- DC supplies and control voltages provided via PCB

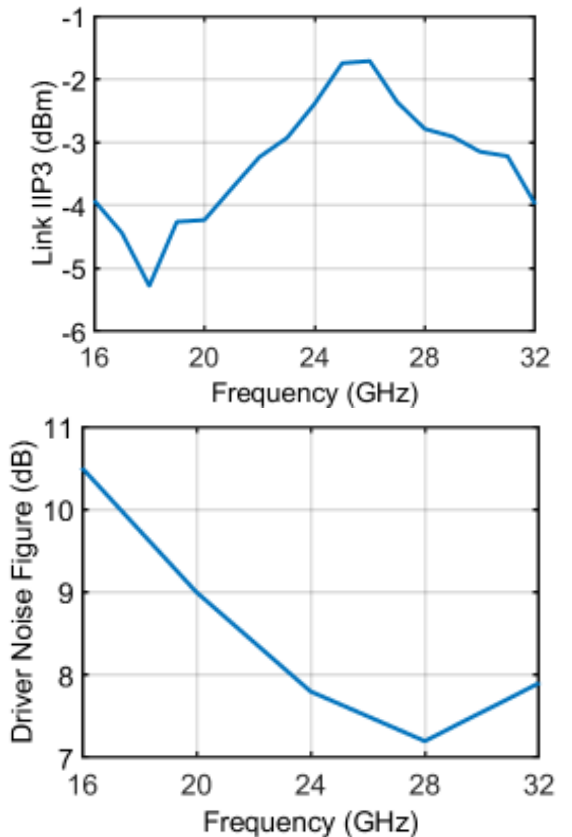
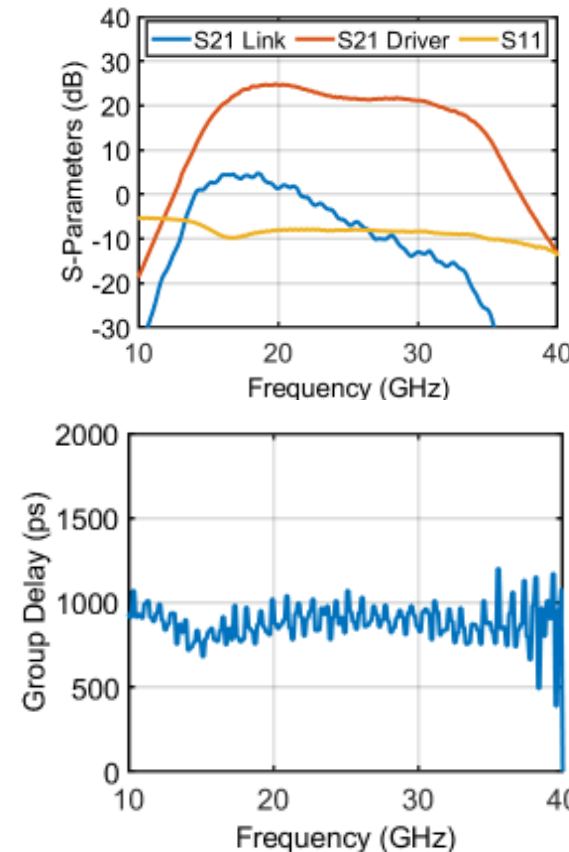
Test Setup



- GSG probe used for RF input
- Vertical grating couplers used to interface with input/output fibers

Measurement Results

- Driver S21 bandwidth 16-32GHz
- S11 lower than -8dB
- Driver NF ranges from 7-10.5 dB
- Link 3dB bandwidth is 15-21GHz due to MZM termination resistance error
- Link group delay variation is 313ps
- Link IIP3 ranges from -5.5 to -1.5dBm



Performance Summary

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Max Voltage Swing	2V _{pp}	N/A	N/A	2.5V _{pp}	2.5V _{pp}
Link IIP3	22dBm	6.8dBm	N/A	4.1dBm	-2dBm
Supply Voltage	2.5/3.3V	1.8V	N/A	0.9/1.6V	0.8/1.6/2.4V
Technique	IM3 Injection	Adaptive Predistortion	Polynomial Predistortion	Predistortion	N/A

*Integrated MZM

RF SiP RX Front-End with Differential MZM Driver Summary

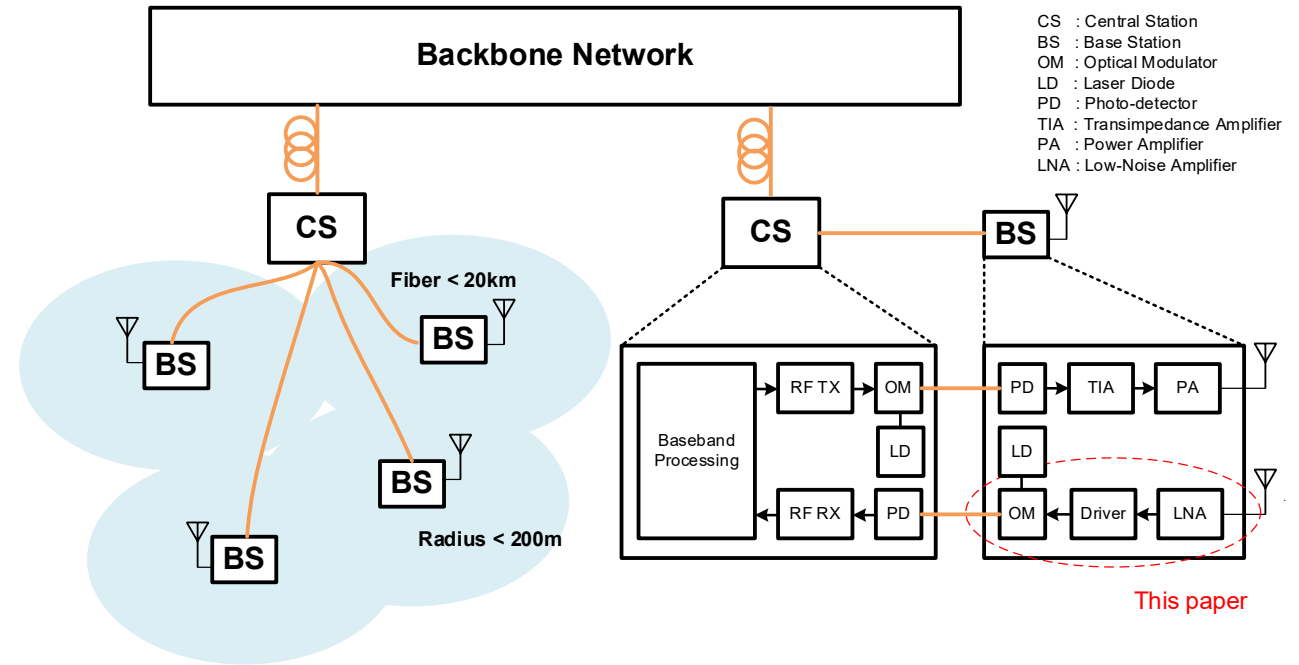
- An integrated wideband photonic receiver front-end is implemented
- 16-32GHz driver bandwidth achieved through co-design with bondwire interconnection and MZM
- CMOS driver adopts stacked-FET structure with MCR technique to achieve wideband performance and high supply voltage operation

Outline

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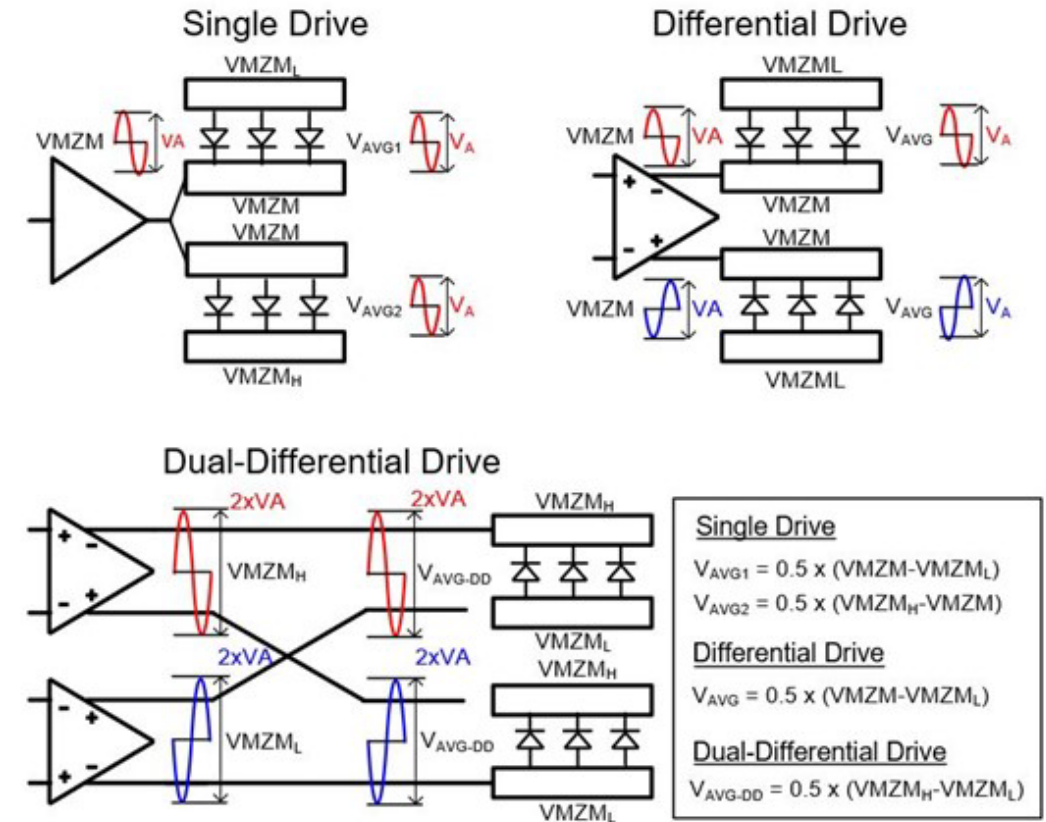
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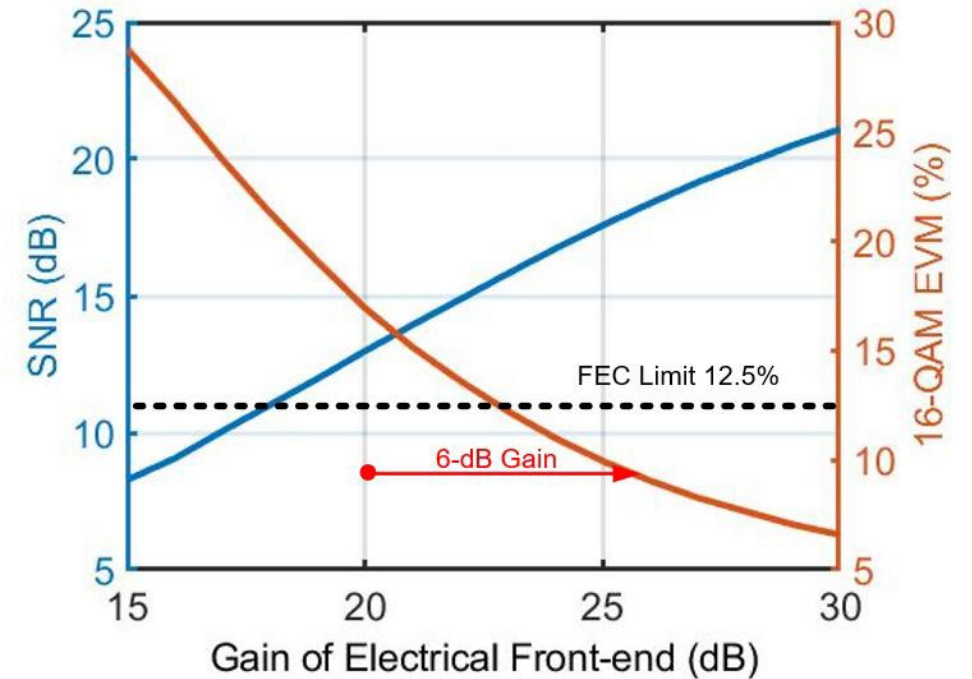
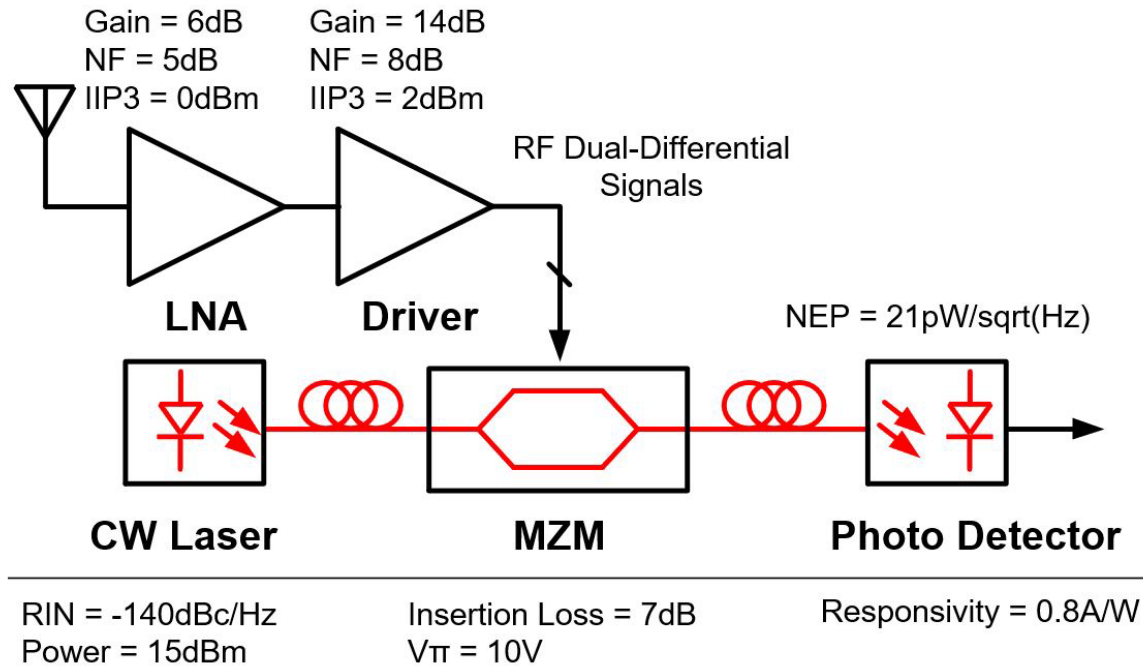


Dual-Differential MZM Driving Scheme

- Differential drive reduces second-order non-linearity compared to single drive
- Higher front-end gain required to achieve better SNR
- Dual-differential drive provides 2X modulation voltage swing, providing 6dB RF gain with same linearity as differential drive

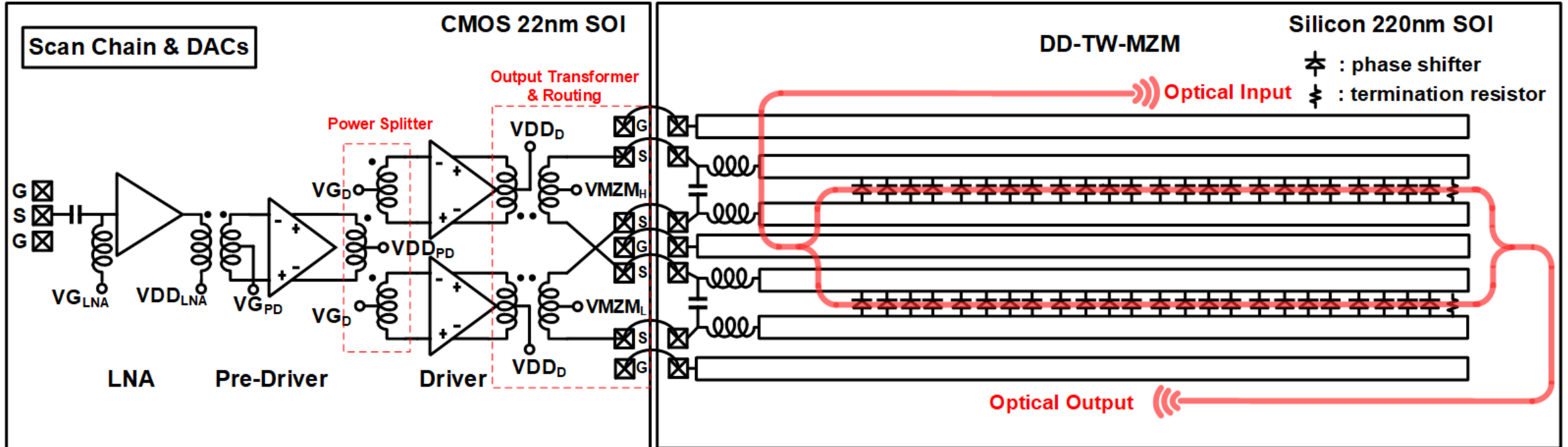


System Analysis



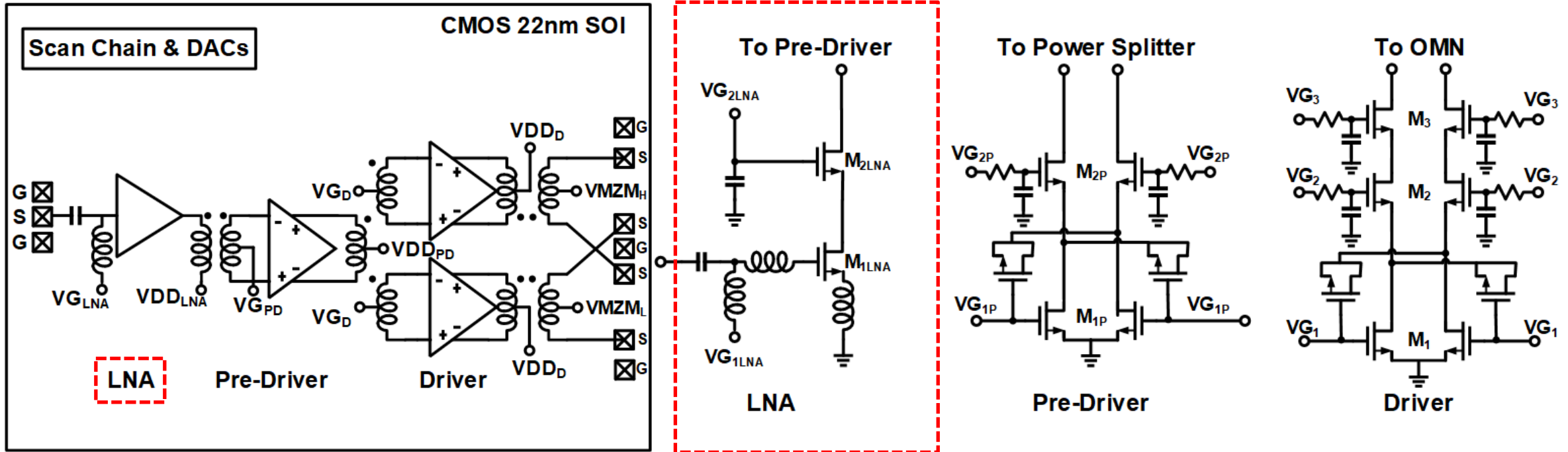
- Higher gain from dual-differential driving scheme translates to higher SNR for required symbol rates

Front-End Architecture



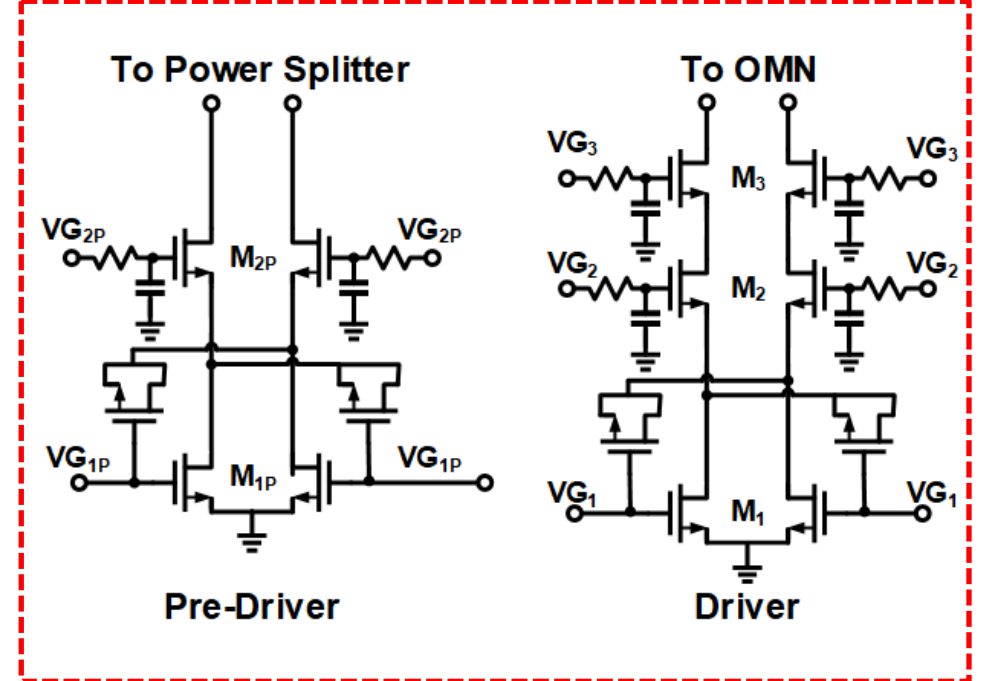
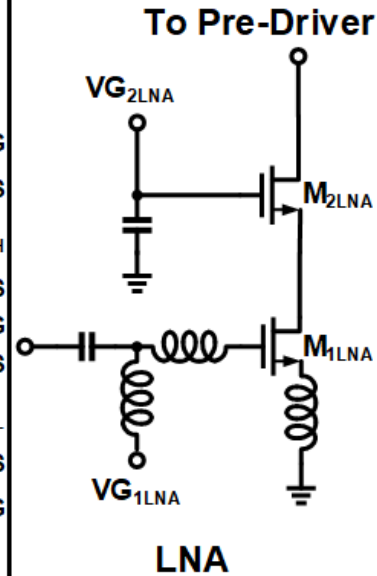
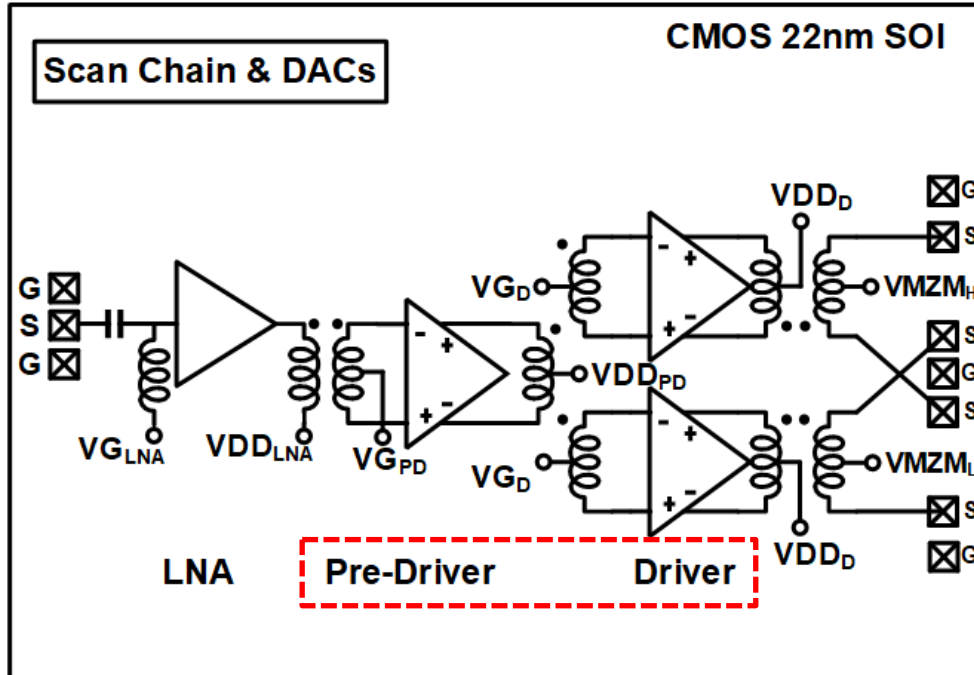
- Power splitter and dedicated output routing is necessary to implement dual-differential driving scheme
- Co-designed LC matching network on silicon PIC

Cascode LNA with Inductive Degeneration



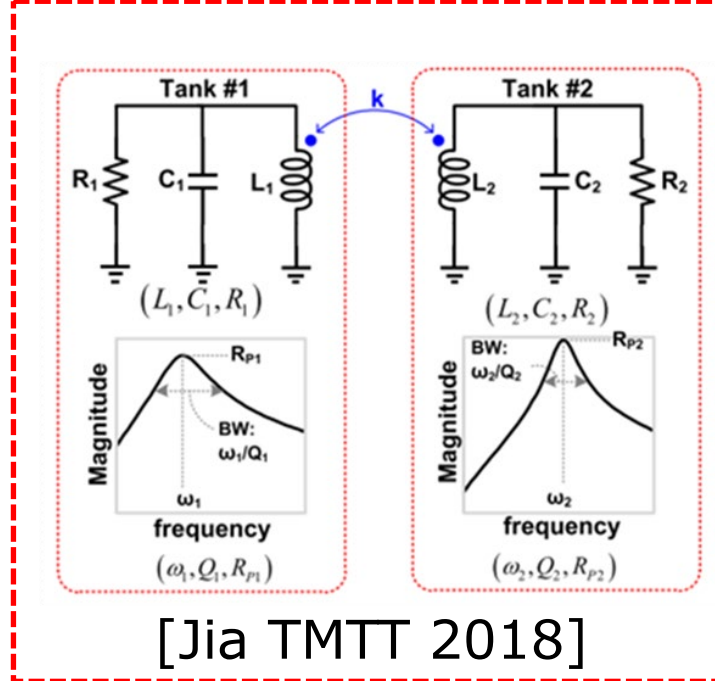
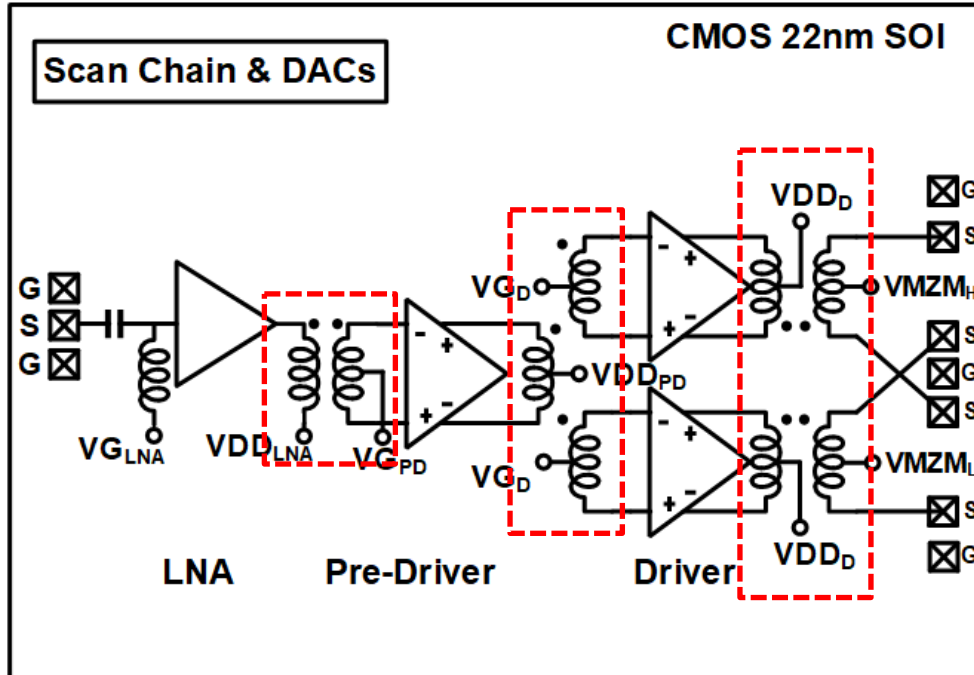
- Cascode structure for high gain and high reverse-isolation
- Inductive degeneration used for input matching

Stacked-FET Pre-Driver and Driver

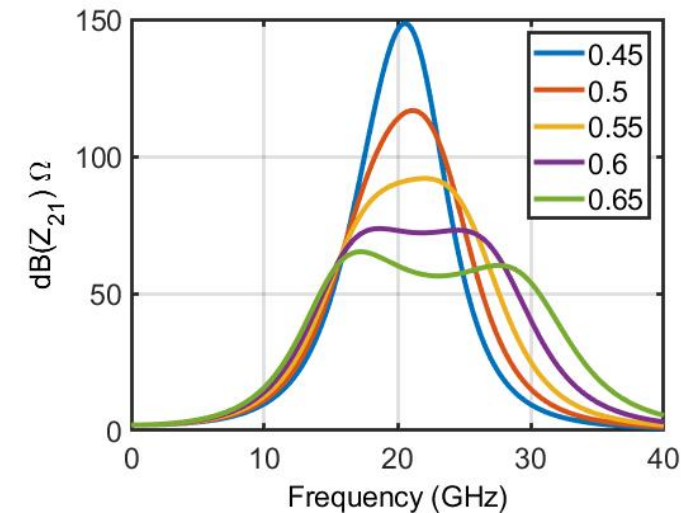


- Stacked-FET structure allows high voltage swing operation without stressing transistors
- Capacitive neutralization improves reverse-isolation

Magnetically-Coupled Resonator Technique

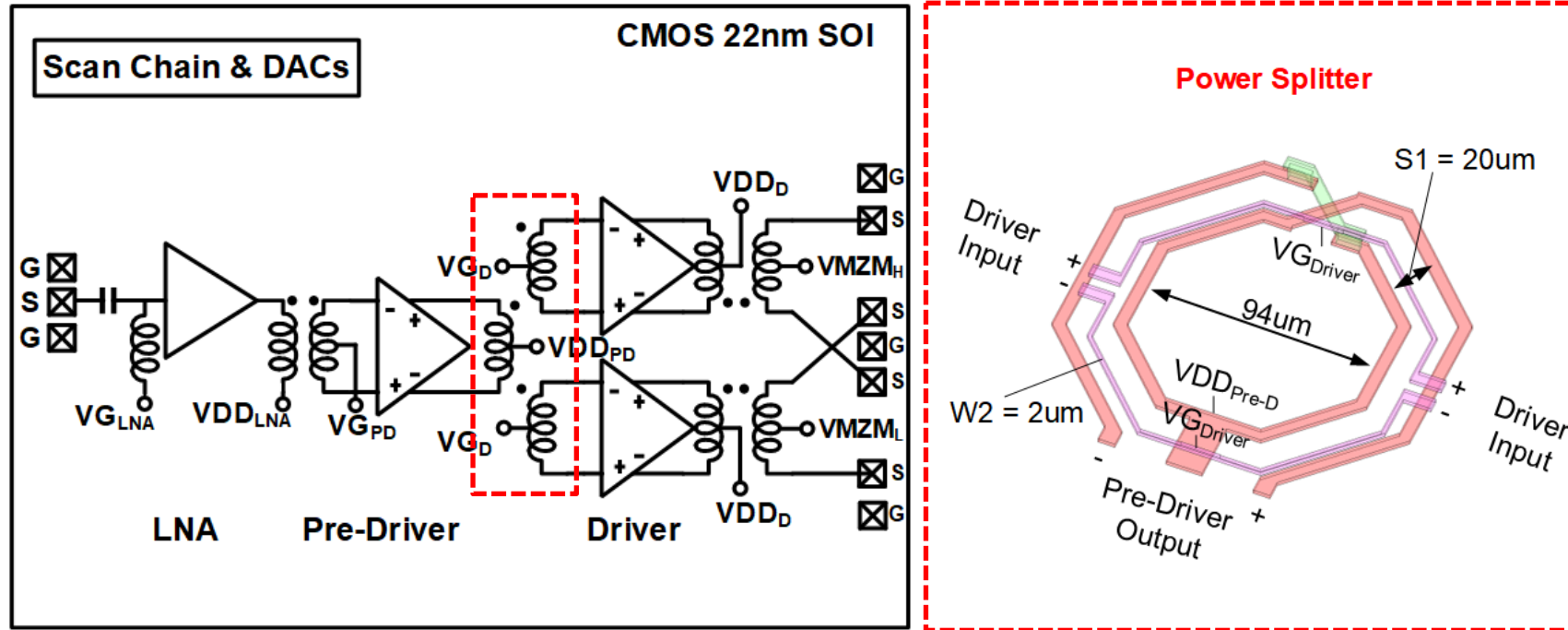


Equivalent Circuit Model Simulation

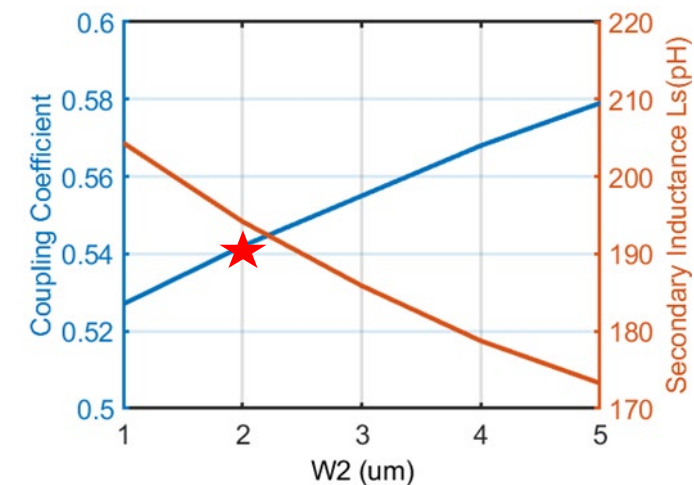
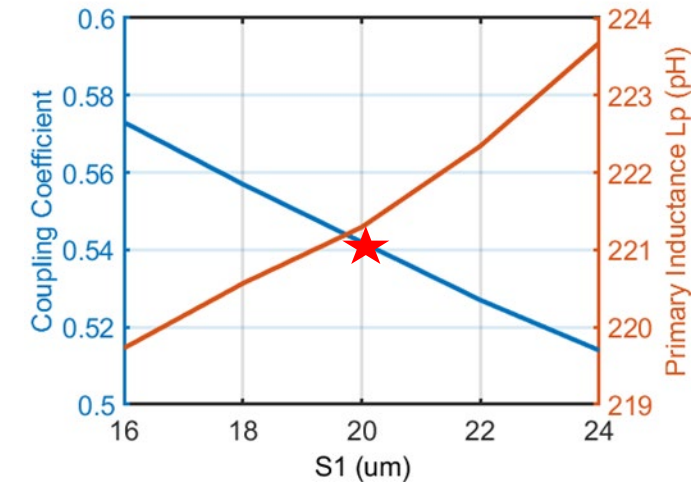


- Used for all inter-stage matching networks in the driver
- Wide-band response by adjusting coupling factor

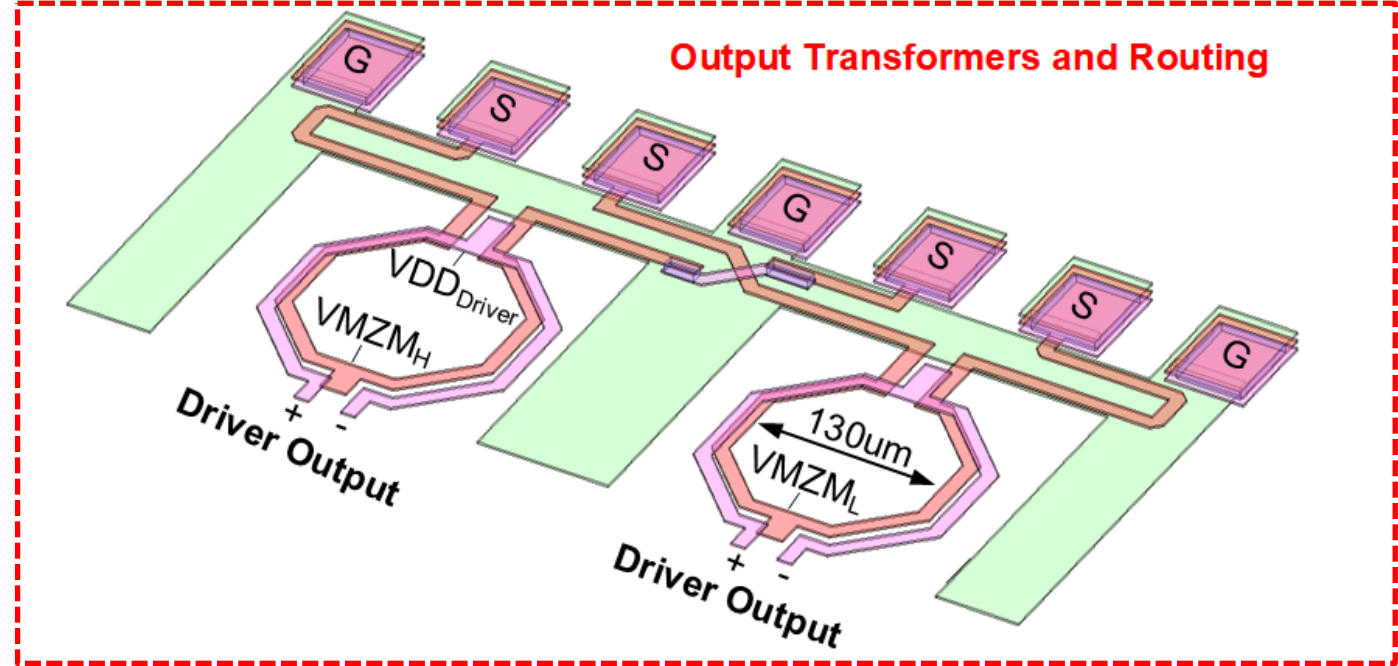
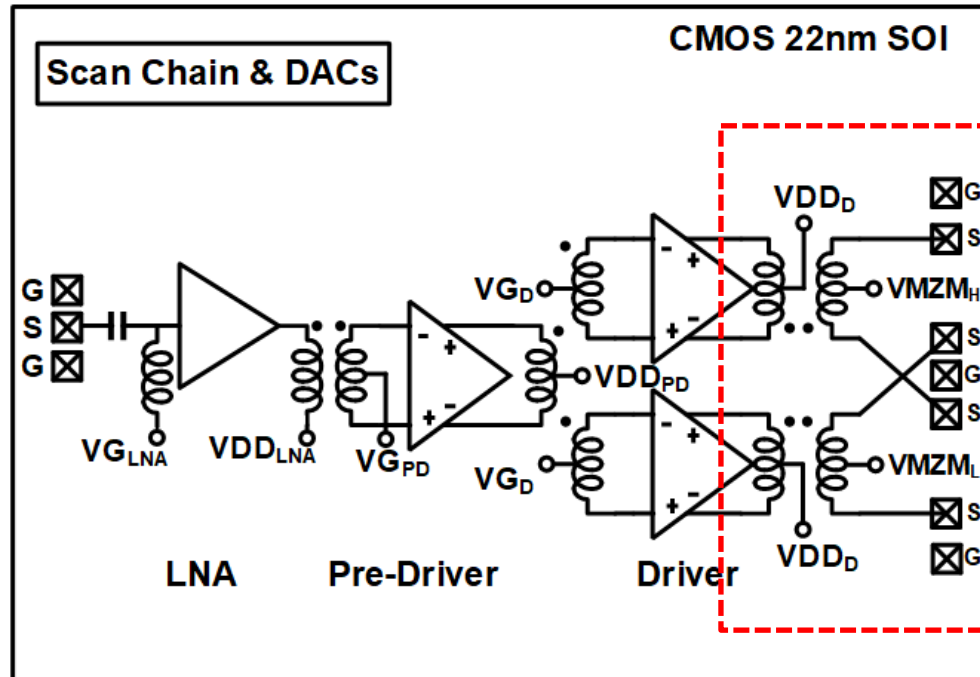
Vertically-Coupled Power Splitter



- Performing power splitting while adopting MCR technique for wide-band response
- Iterative optimization of W2 and S1 for optimal coupling coefficient

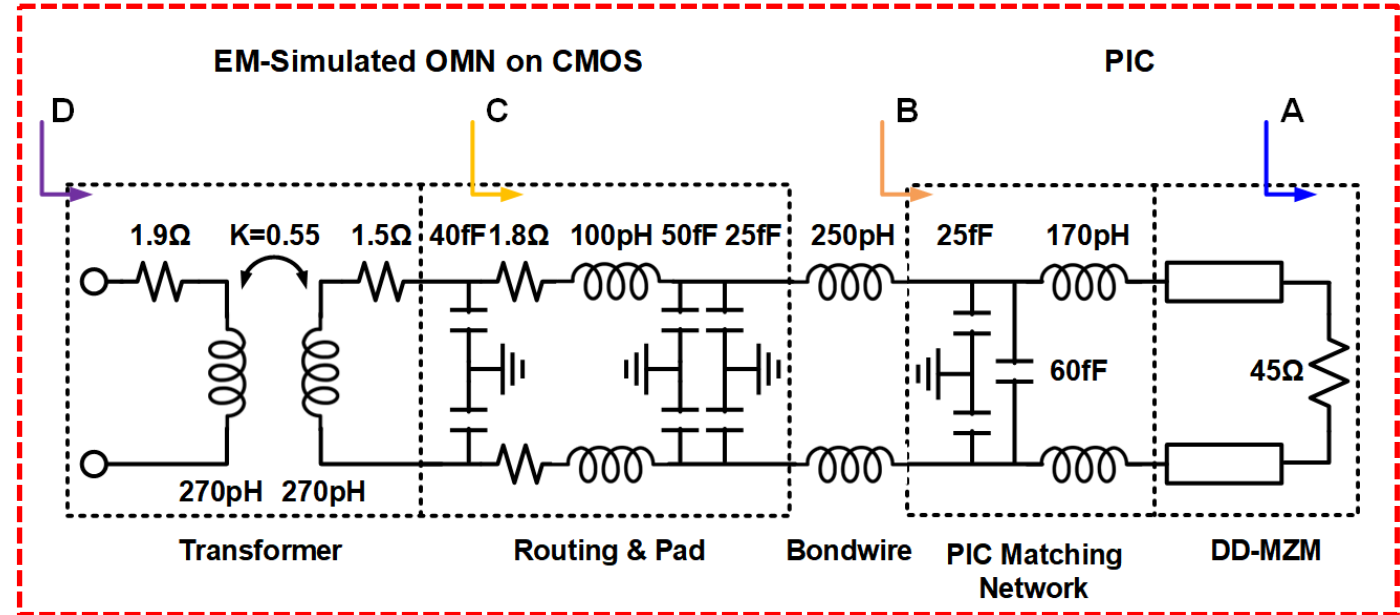
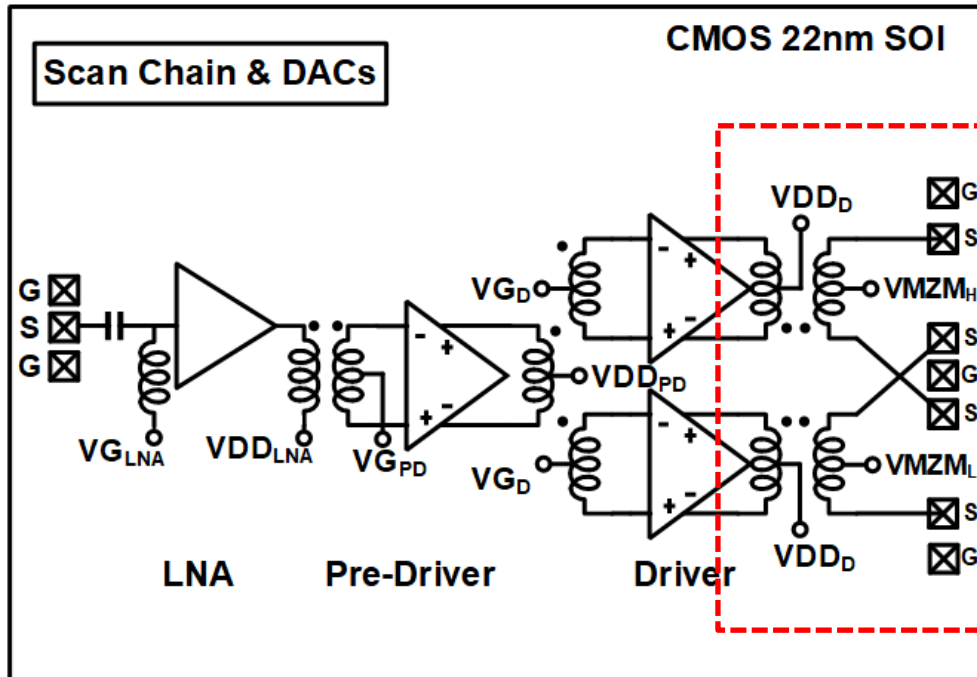


CMOS Output Matching Network

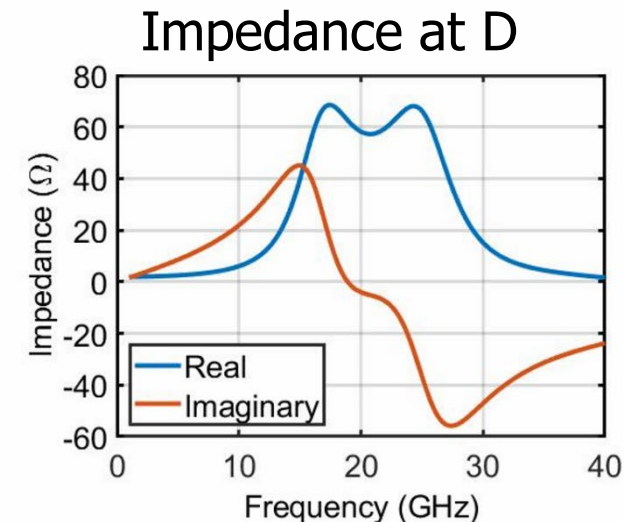
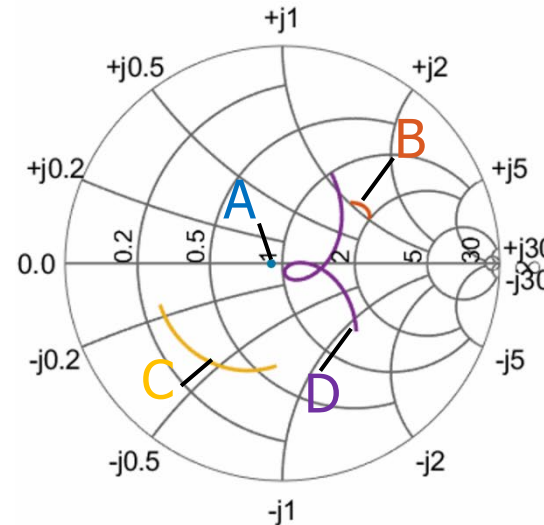


- Phase-compensated routing for dual-differential drive
- MZM reverse-bias voltage supplied via transformer center-tap

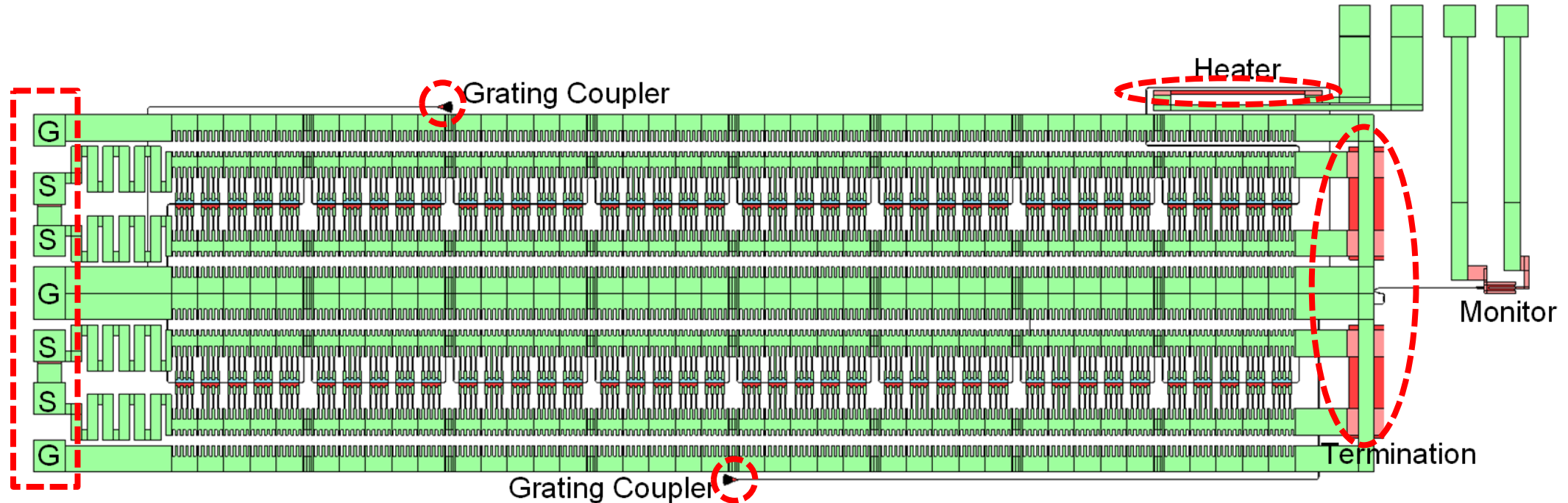
CMOS Output – PIC LC Matching Network



- Impedance transformation from MZM termination to driver impedance

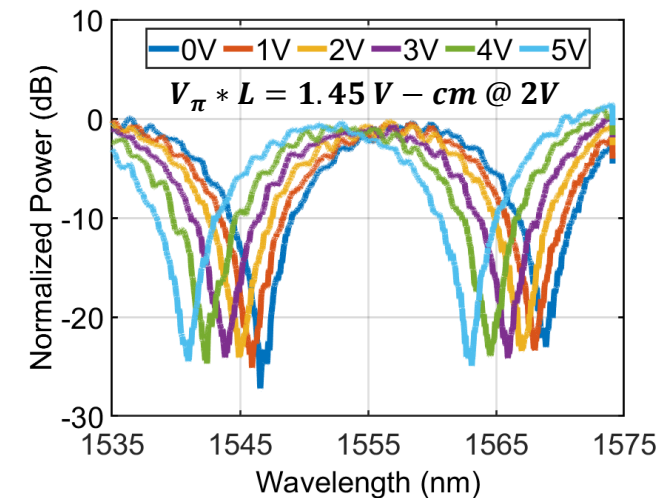
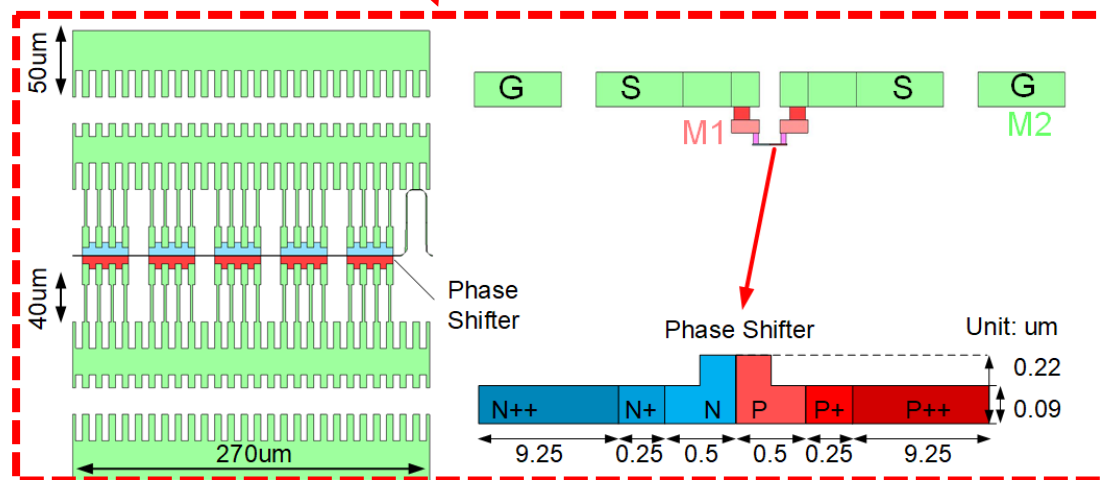
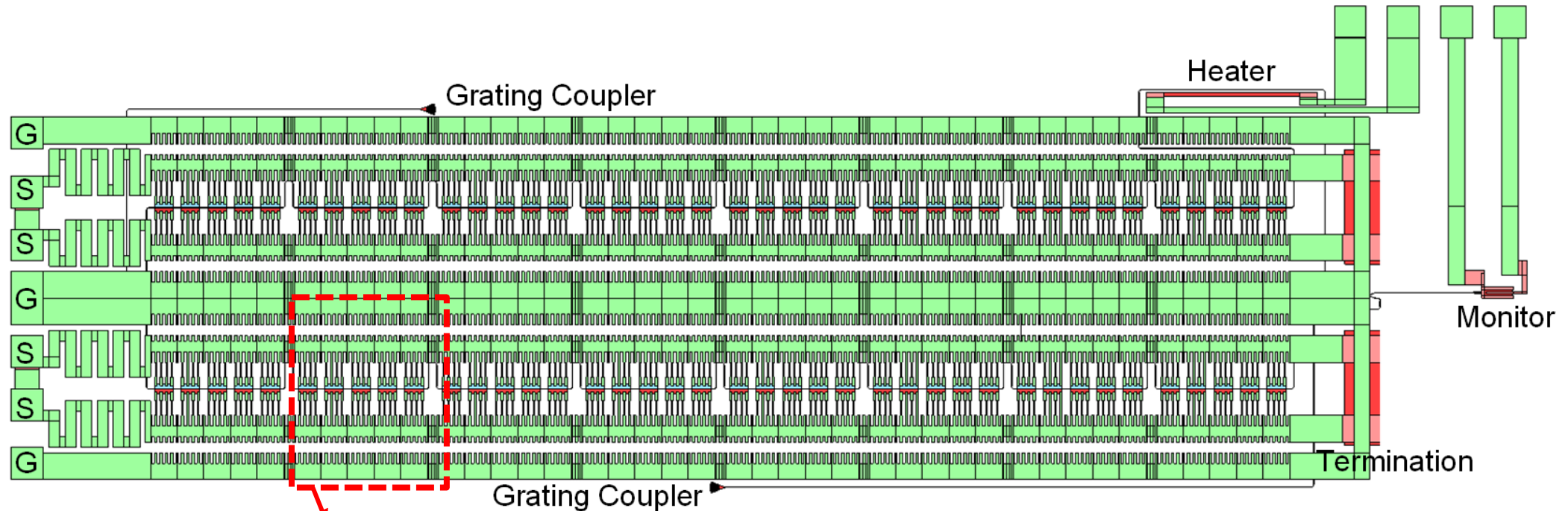


Dual-Differential Traveling-Wave MZM

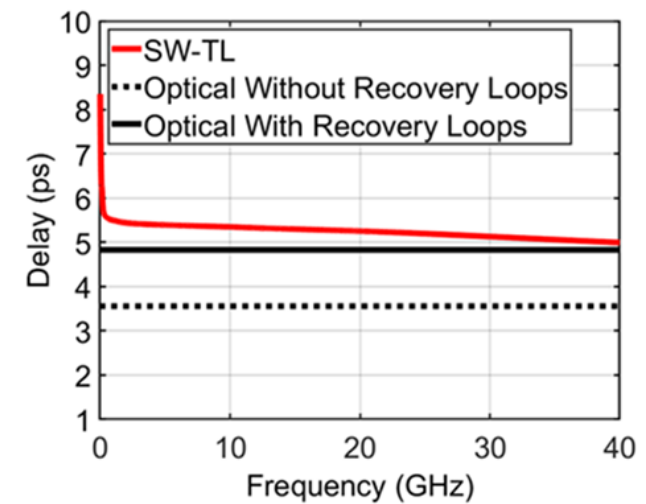
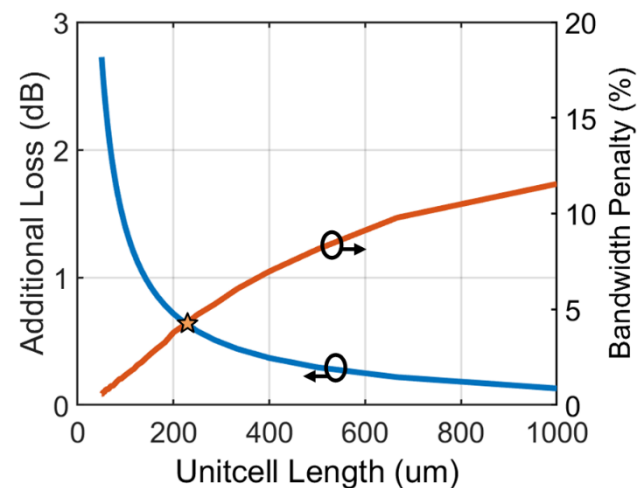
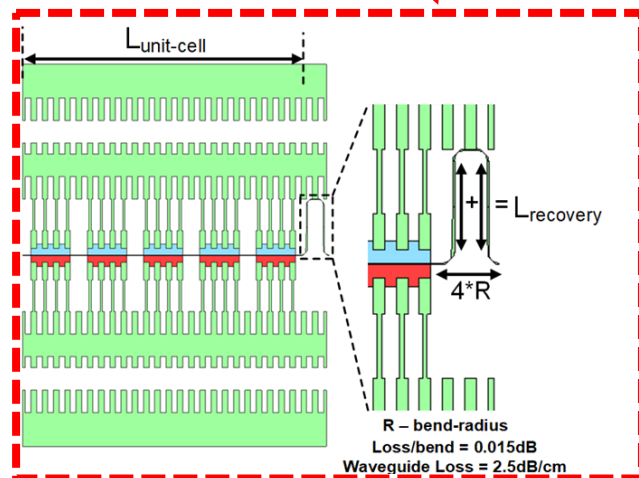
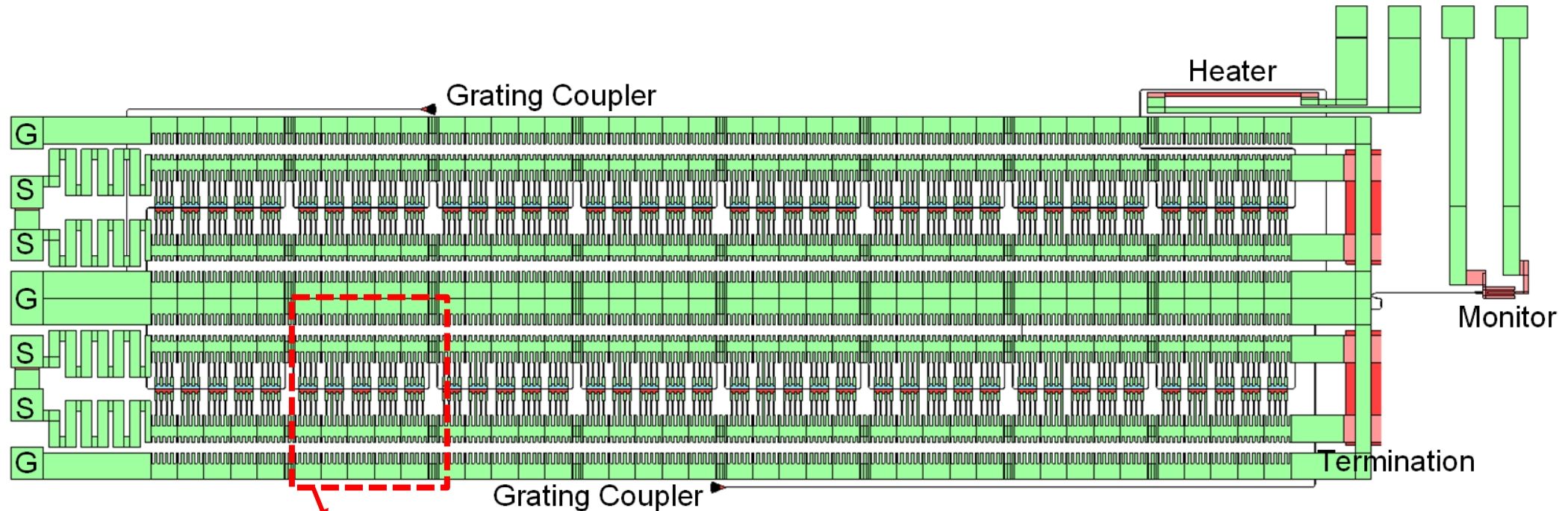


- GSSGSSG electrode configuration with on-chip termination
- Vertical grating couplers for optical input and output
- Thermal phase shifter (heater) for bias control

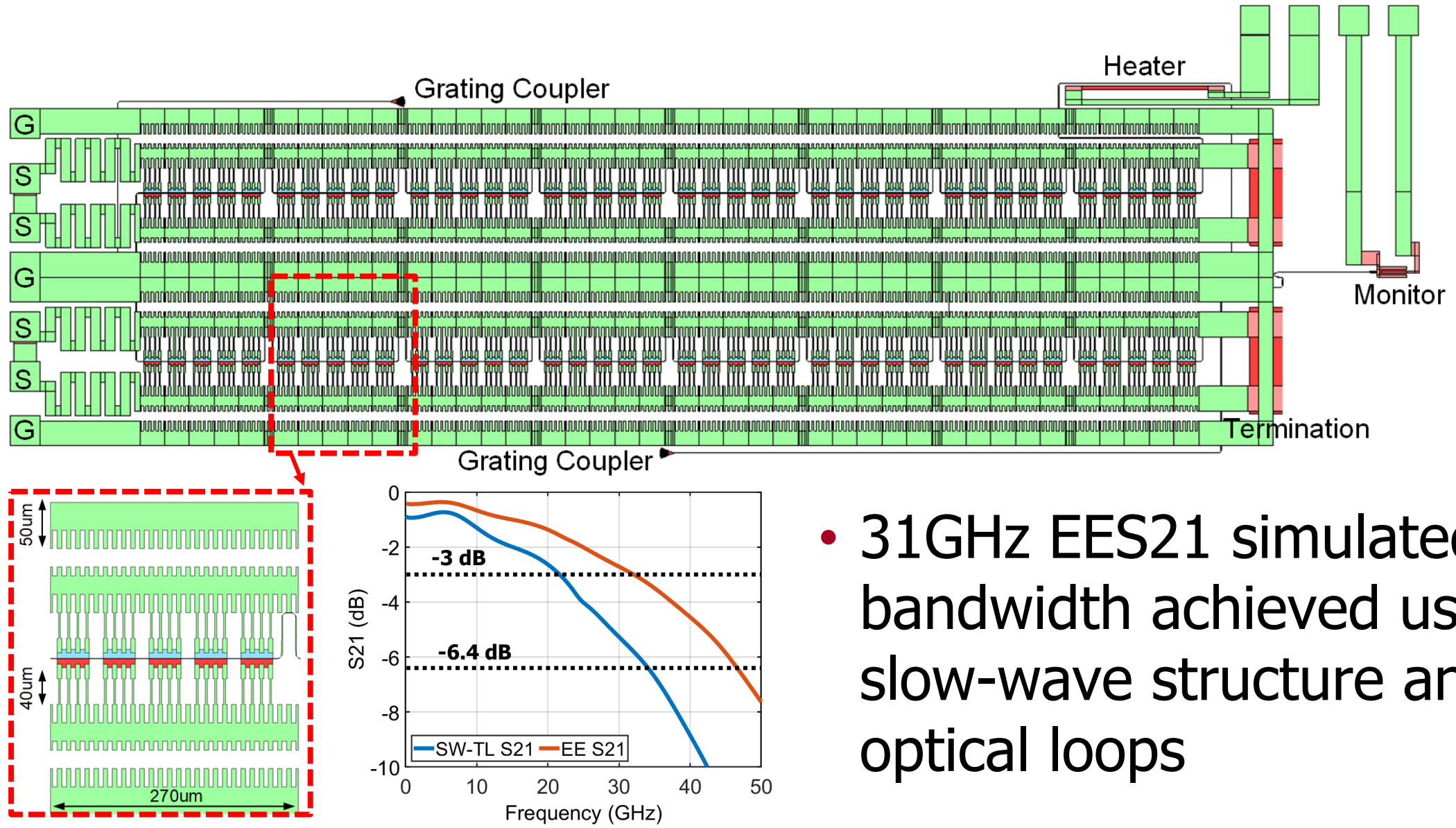
Lateral PN Junction Depletion-Mode Phase Shifter



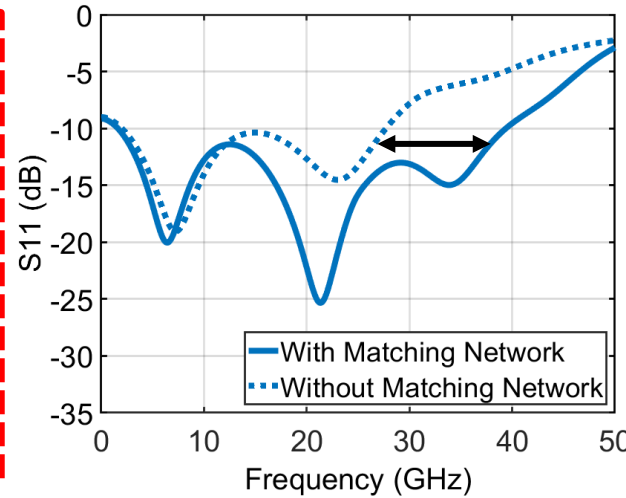
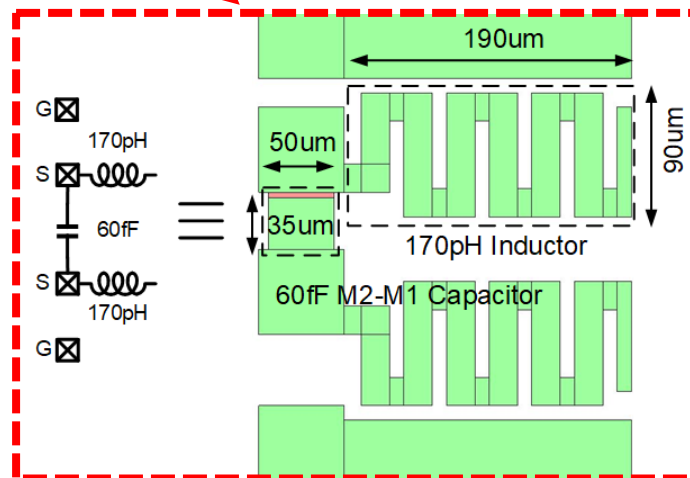
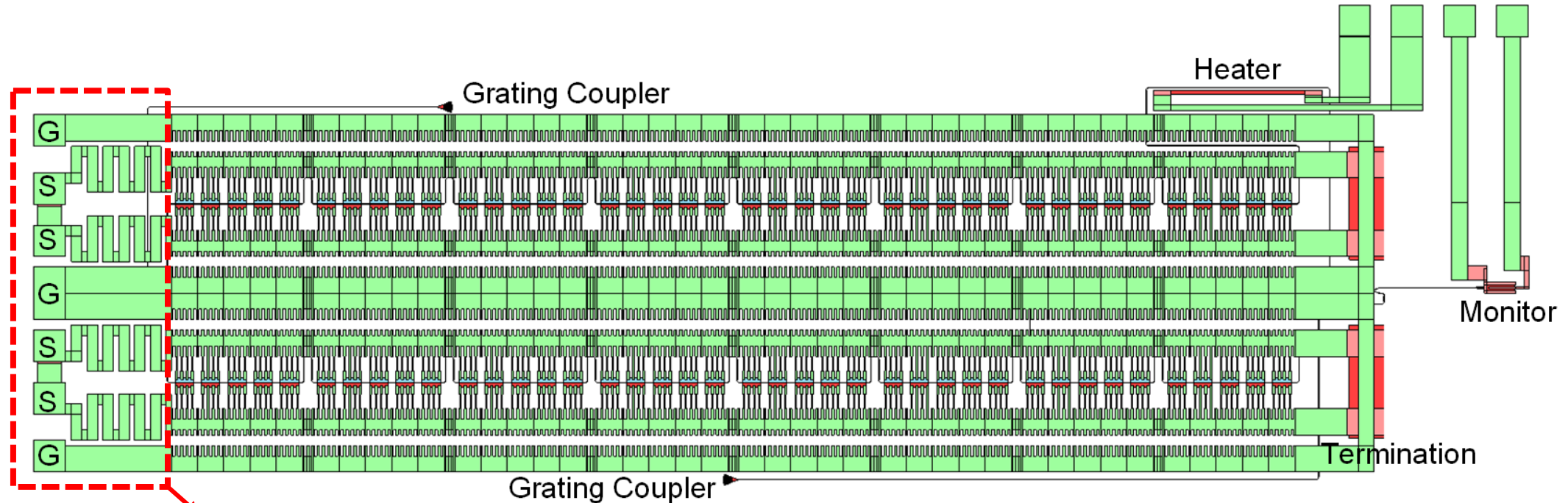
Optical Recovery Loops



Slow-Wave Differential Transmission Line

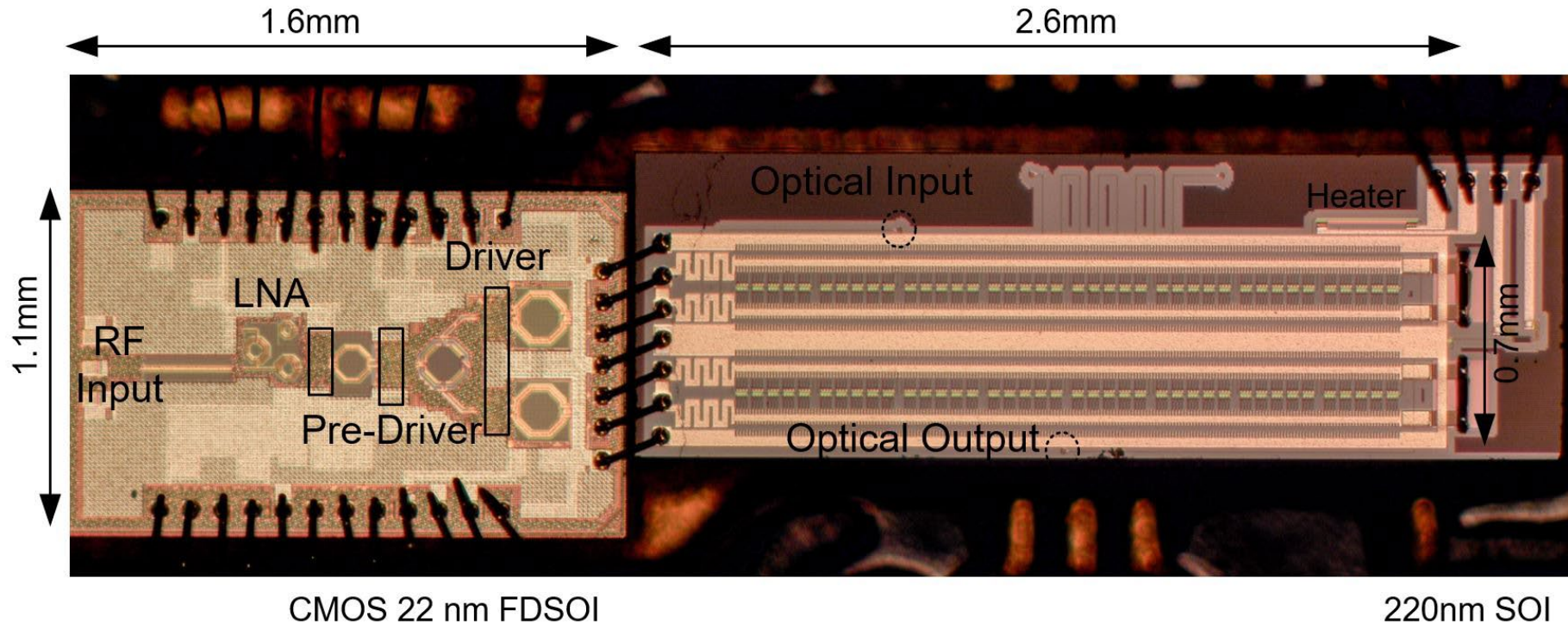


PIC LC Matching Network



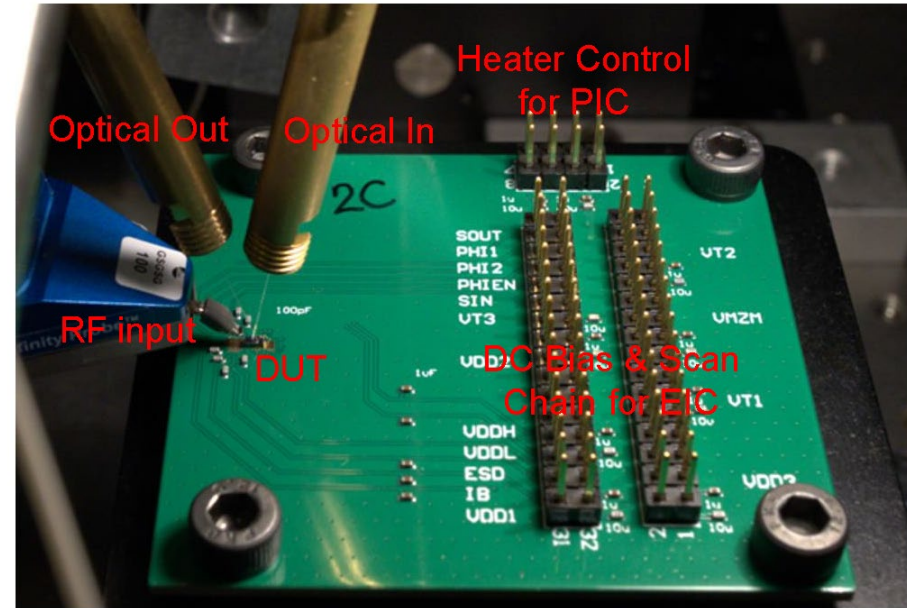
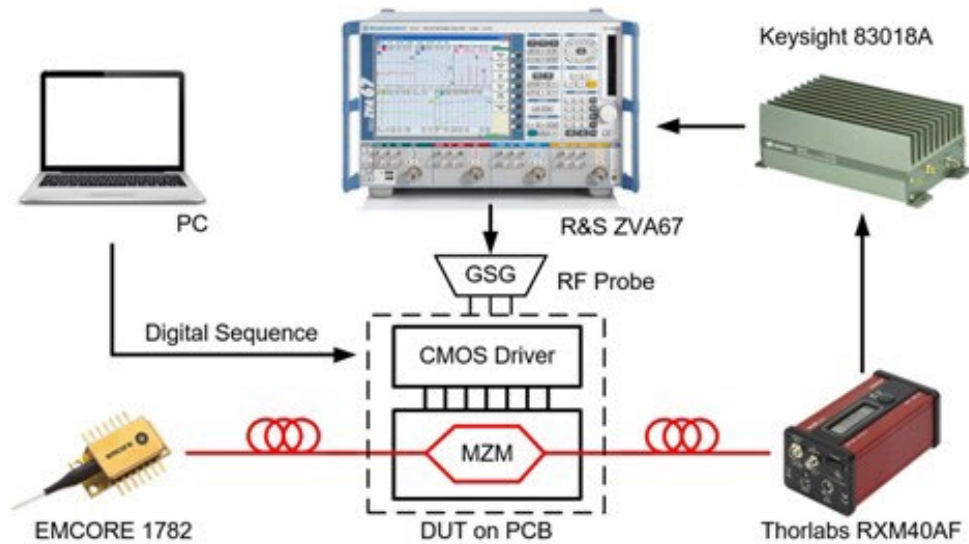
- Improved matching between 15-25GHz and S_{11} bandwidth extension to 40GHz

Integrated Silicon RoF Front-End



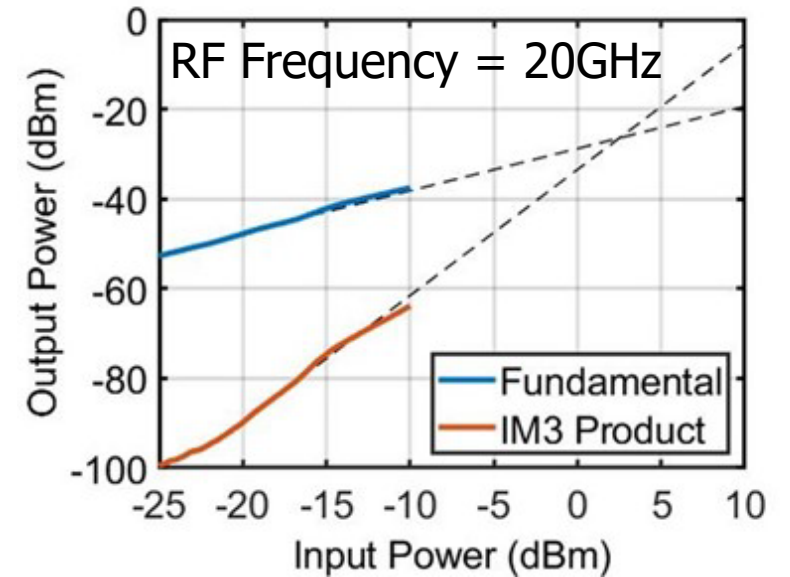
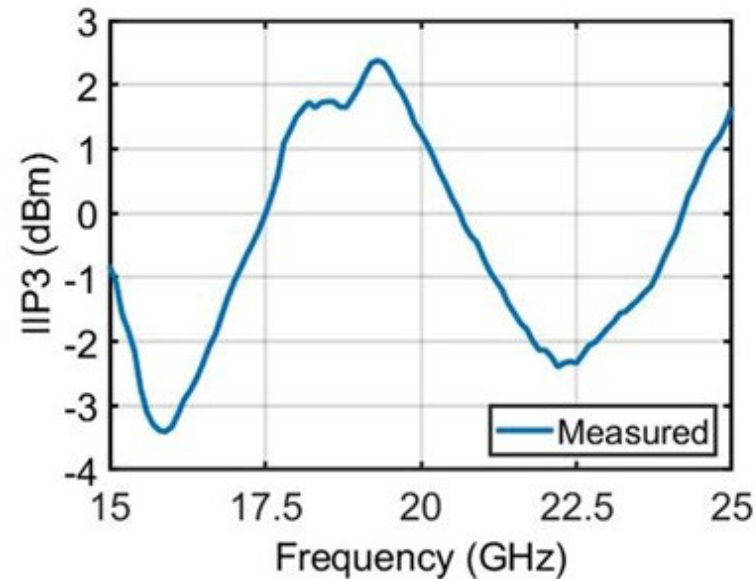
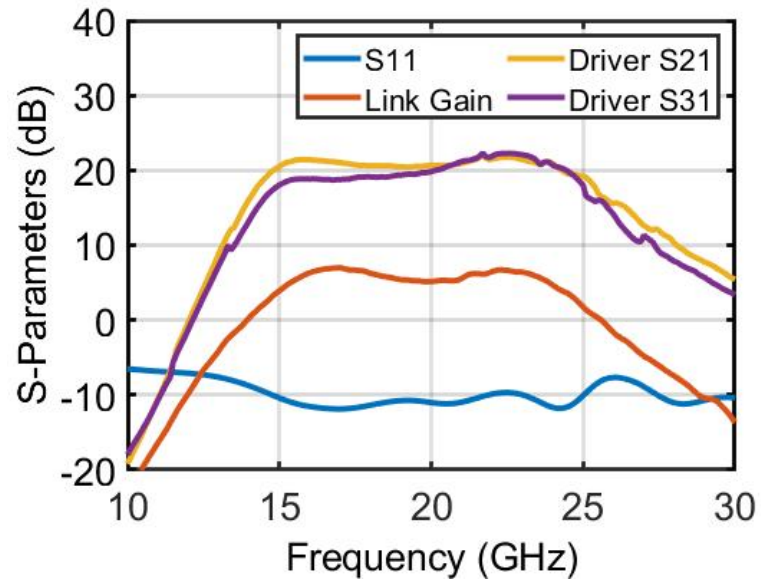
- Driver and PIC mounted on PCB with minimal gap to reduce bondwire inductance

Bandwidth and Linearity Measurement Set-up



- Enables s-parameter and inter-modulation distortion characterization
- RF input to driver applied using GSG probe and PIC optical input/output with vertical fiber grating couplers

Bandwidth and Linearity Measurement Results



- 15-25GHz link bandwidth with 6dB gain and $S_{11} < -10\text{dB}$
- 2.2dB maximum IIP3 in operational bandwidth

Performance Summary

References	[Zhu JLT 2016]	[Hosseinzadeh JSSC 2021]	[Hosseinzadeh MTT 2019]	[Luo JLT 2023]	[Luo JLT 2024]	This Work
Frequency	2-30	0.5-20	1-20	20-35	16-32	15-25
Process (Driver/MZM)	– /LiNbO3	SiGe/Si	SiGe/Si	CMOS/LiNbO3	CMOS/Si	CMOS/Si
Integration	Discrete	Hybrid	Hybrid	Discrete	Hybrid	Hybrid
Power Consumption (mW)	N/A	1700	1650	180	209	448
Modulation Voltage Swing (V _{pp})	N/A	2	3	2.5	2.5	5
IIP3 (dBm)	6.2	22	6.8	4.8/-15	1	2
Supply Voltage (V)	N/A	3.3	N/A	0.9/1.6	0.8/1.6/2.4	0.8/1.6/2.4
NF	N/A	14**	14**	22/6	38.1*/13.5**	34.1*/13.4**
SFDR	111	120**	109**	99/101	89*/105**	92*/106**
Overall Gain (dB)	N/A	0	0	5	0	6

* Grating Coupler

** Edge Coupler

RF SiP RX Front-End w/ Dual-Differential MZM Driver Summary

- Presented the co-design and implementation of a novel dual-differential driving scheme-based Radio-Over-Fiber front-end
- CMOS driver with compact power splitter enables dual-differential drive
- Output matching with LC network on PIC allows for bondwire-based integration with broadband operation
- Dual-differential MZM implemented using slow-wave techniques and optical recovery loops
- System achieves 15-25 GHz bandwidth with 6dB gain and 2GBps 16-QAM (8Gbps) operation